RQF LEVEL 3



NITIS301

NETWORKING AND INTERNET TECHNOLOGIES

IoT System Installation

TRAINER'S MANUAL

October, 2024

IoT SYSTEM INSTALLATION

KOICA Korea International Cooperation Agency

TQUM TVET Quality Management Project

2024

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Original published version: October, 2024

ACKNOWLEDGEMENTS

The publisher would like to thank the following for their assistance in the elaboration of this training manual:

Rwanda TVET Board (RTB) extends its appreciation to all parties who contributed to the development of the trainer's and trainee's manuals for the TVET Certificate III in Networking and Internet Technologies, specifically for the module **" NITIS301: IoT System Installation."**

We extend our gratitude to KOICA Rwanda for its contribution to the development of these training manuals and for its ongoing support of the TVET system in Rwanda

We extend our gratitude to the TQUM Project for its financial and technical support in the development of these training manuals.

We would also like to acknowledge the valuable contributions of all TVET trainers and industry practitioners in the development of this training manual.

The management of Rwanda TVET Board extends its appreciation to both its staff and the staff of the TQUM Project for their efforts in coordinating these activities.

This training manual was developed:

Under Rwanda TVET Board (RTB) guiding policies and directives

Under Financial and Technical support

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ACRONYMS

5G: 5th Generation

ACK: Acknowledgement

ARIB: Association of Radio Industries & Businesses

AWS: Amazon Web Services

BLE: Bluetooth Low Energy

CIOT: Cellular Internet of Things

CI: Coding Indicator (for Bluetooth 5)

CI: Connection Interval (BLE)

CE: Coverage Enhancement

CoAP: stands for Constrained Application Protocol

DHT: Digital Temperature and Humidity

EDGE: Enhanced Data Rates for GSM Evolution

EtherCAT: Ethernet for Controlled Automation Technology

EC-GSM-IoT: Extended Coverage GSM Internet of Things

FDD: Frequency Division Duplexing

FSK: Frequency Shift Keying

FreeRTOS: Free Real-Time Operating System

GPRS: General Packet Radio Service

GSM: Global System for Mobile Communications

GND: Ground

HTTP: Hypertext Transfer Protocol

HTTPS: Hypertext Transfer Protocol Secure

IIOT: Industrial Internet of Things

IIoT: Infrastructure IoT

IoMT: Internet of Military Things

IoT: Internet of Things

ISM: Industrial, Science, and Medicine

IP: Internet Protocol

IoT: Internet of Things

ISP: Internet Service Provider

JTAG: Joint Test Action Group

LEDs: Light Emitting Diodes

LoRa: Long Range Radio

LPWAN: Low-power Wide Area Network

LTE: Long-Term Evolution

LTE-M LTE for Machine-Type Communications

M2M: Machine to Machine

MCU: Microcontroller unit

MTC: Machine Type Communications

MQTT: Message Queuing Telemetry Transport

NB-IoT: Narrowband Internet of Things

NB-PLC: Narrow-Band PLC

NFC: Near-Field Communication

PAN: Personal Area Network

PRTG: Paessler Router Traffic Grapher

PLC: Power Line Communication

PCP: priority Code Point

Profinet: Process Field Network

PCG: Project Coordination Group

PDU: Protocol Date Unit

QoS: Quality of Service

RAN: Radio Access Networks

RTB: Rwanda TVET Board

TVET: Technical and Vocational Education and Training

TQUM: TVET Quality Management Project

RTOS: Real-Time Operating System

RU: Resource Unit

RFID: Radio Frequency Identification

RX: Receive

SNR: Signal-to-Noise Ratio

TbE: TeraBit Ethernet

TVET: Technical and Vocational Education and Training

TQUM: TVET Quality Management Project

SDR: Software-defined radio

TCP/IP: Transmission Control Protocol/Internet Protocol

TX: Transmit

VLAN: Virtual Local Area Network

Wi-Fi: Wireless Fidelity

INTRODUCTION

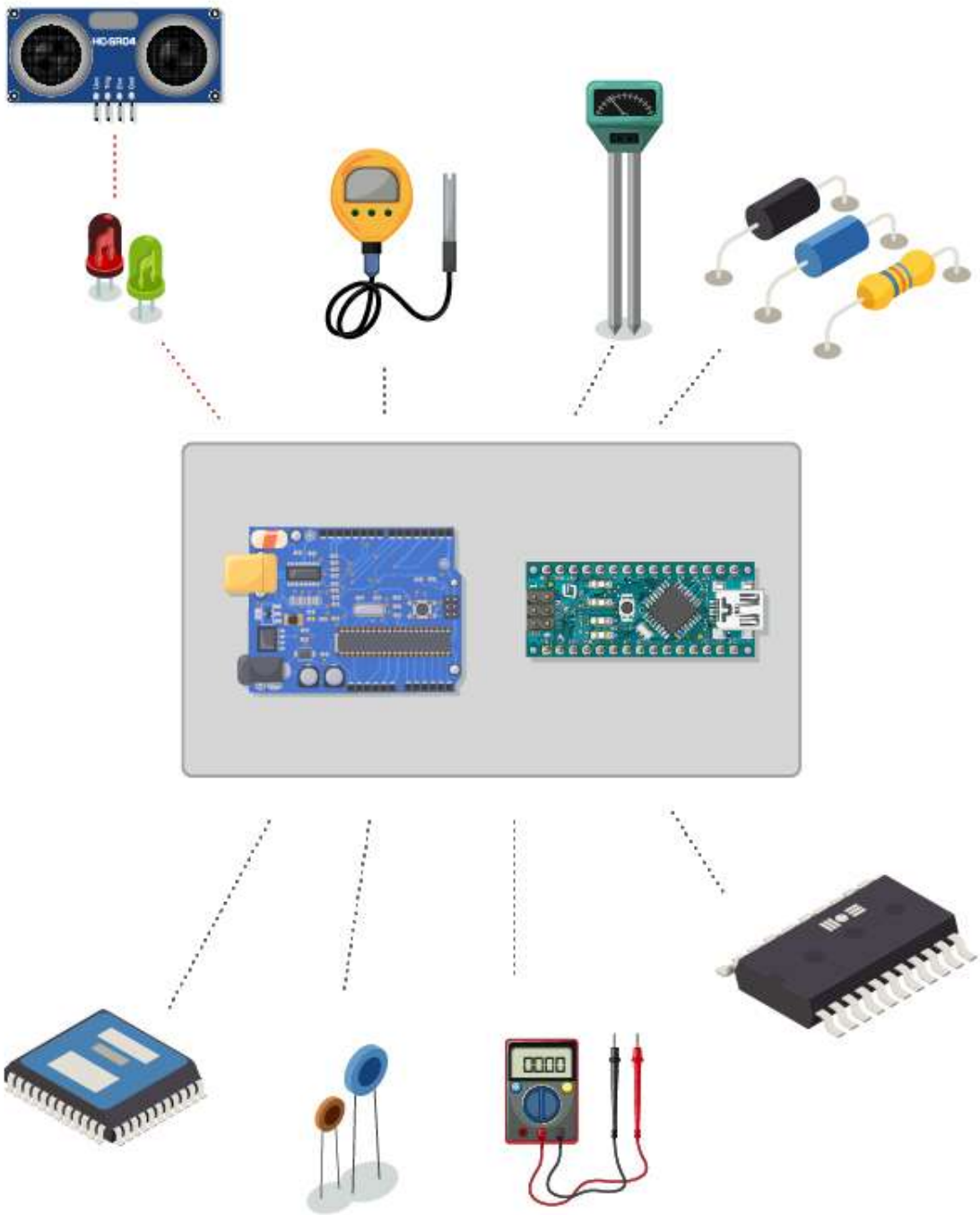
This trainer's manual includes all the methodologies required to effectively deliver the module titled **"IoT system Installation."** Trainees enrolled in this module will engage in practical activities designed to develop and enhance their competencies.
The development of this training manual followed the Competency-Based Training and Assessment (CBT/A) approach, offering ample practical opportunities that mirror real-life situations.

The trainer's manual is subdivided into Learning Outcomes, each learning outcome has got various topics, you will start guiding a self-assessment exercise to help students rate themselves on their level of skills, knowledge and attitudes about the unit. The trainer's manual will give you the information about the objectives, learning hours, didactic materials, proposed methodologies and crosscutting issues.
A discovery activity is followed to help students discover what they already know about the unit.

This manual will give you tips, methodologies and techniques about how to facilitate students to undertake different activities as proposed in their trainee's manuals. The activities in this training manual are prepared such that they give opportunities to students to work individually and in groups. After going through all activities, you shall help students to undertake progressive assessments known as formative and finally facilitate them to do their self-reflection to identify your strengths, weaknesses and areas for improvements. Remind them to read the point to remember section which provides the overall key points and takeaways of the unit.

MODULE CODE AND TITLE: NITIS301 IoT SYSTEM INSTALLATION

Learning Outcome 1: Prepare IoT system installation
Learning Outcome 2: Deploy IoT equipment
Learning Outcome 3: Operate IoT system
Learning Outcome 4: Test IoT system
Learning Outcome 5: Maintain IoT system



Indicative Contents

1.1 Introduction to IoT

1.2 Identification of IoT Network Technologies

1.3 Description of IoT Architecture and Protocols

1.4 Calibration of Tools, Equipment and Materials

1.5 Interpretation of Manual/Datasheet

Key Competencies for Learning Outcome 1: Prepare IoT System Installation

Knowledge	Skills	Attitudes
• Description of IoT fundamentals • Description of IoT Components • Identification of IoT Network Technology and their Challenges • Identification of IoT Tools, Materials, and Equipment • Description of IoT Architecture and Protocols • Description of Manual and Datasheet	• Selecting IoT Network Technologies • Interpreting Manual and Datasheet • Calibrating IoT Tools and Equipment	• Having an Innovation • Having Team Work • Having Time Management • Having Creativity • Having Self Confidence

Duration: 20hrs

Learning outcome 1 objectives:

By the end of the learning outcome, the trainees will be able to:

1. Describe correctly IoT fundamentals based on IoT functionalities and applications
2. Describe correctly IoT component as used in IoT system installation
3. Select correctly IoT tools, materials and equipment as used in IoT system installation
4. Identify properly IoT network technology and challenges based on IoT system installation
5. Calibrate accurately IoT Tool, devices, and equipment as per manufacturer's standards
6. Interpret properly Installation manual based on IoT system installation

Resources

Equipment	Tools	Materials
• Wireless Router • Gadgets • Appliances • Computer	• Calibration software • Simulators • Screw drivers' kit • Pliers set • Digital multimeter	• IoT Device • Internet • Arduino Starter Kit, • IoT Device Datasheet

Advance Preparation:

Before delivering this learning outcome, you are recommended to:

- Avail designed IoT system and its installation manual
- Avail resources for participants to further explore IoT system installation techniques and best practice

Indicative content 1.1: Introduction to IoT

Duration: 4 hrs

Theoretical Activity 1.1.1: Description of IoT Fundamentals

Notes to the trainer:

- Trainer may use small group to describe IoT fundamentals, the use of images and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and ask learners to answer the following questions:

i. What is the definition of IoT?
ii. Name a few key benefits and challenges of implementing IoT in various industries
iii. Provide examples of real-world applications where IoT technology is extensive ly used
iv. What are some common devices or components that are essential for building an IoT system?
v. Explain the role of IoT platforms in the development and management of IoT s olutions.
vi. How has IoT technology evolved over the years, and what are some significan t milestones in its development?

Step 2: Asks trainees to write provided answers on flipchart/paper.

Step 3: Asks trainees to present their findings to the whole class.

Step 4: Provides expert view and Address any questions or concerns.

Step 5: Ask trainees to read the key reading 1.1.1 in the trainee manual.



Points to Remember

- **IoT** connects everyday objects to the internet for data exchange and interaction. Devices communicate autonomously, forming interconnected networks
- **Challenges Facing the IoT Industry** are Concerns over security, Interoperability, Scalability and Standardization.
- **Benefits of Implementing IoT** are Enhances efficiency, data analysis, and decision-making, offers cost savings through predictive maintenance and optimization and Improves convenience and quality of life with smart devices.
- **IoT** used in smart home automation**,** Industrial processes and Healthcare, and agriculture.
- **Common Devices** Includes sensors, actuators, microcontrollers/microprocessors and gateway
- **Common components** includes connectivity modules, power source, operating systems, cloud platform and applications
- **Role of IoT Platforms** is to facilitate development, deployment, and management of IoT solutions and Manage device connectivity, data handling, and security features.
- **Evolution of IoT Technology and Milestones** progressed from basic communication to complex device ecosystems. Major milestones include IPv6 adoption and the rise of edge computing.



Theoretical Activity 1.1.2: Identification of IoT components



Notes to the trainer:

- Trainer may use small group for Identification of IoT components the use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. What are "Things" in the context of IoT, and can you give two examples?
ii. How does the "Internet" component support IoT devices?

iii. Name two wireless technologies used for IoT connectivity

Step 2: Asks trainees to write provided answers on flipchart/paper.

Step 3: Asks trainees to present their findings to the whole class.

Step 4: Provides expert view and Address any questions or concerns.

Step 5: Ask trainees to read the key reading 1.1.2 in the trainee manual.



Points to Remember

- **Things:** Physical objects with sensors, actuators, and embedded technologies.
- **Internet:** The global network infrastructure, cloud services, and data management systems.
- **Connectivity:** The communication technologies and protocols that enable data exchange.



Theoretical Activity 1.1.3: Identification of IoT Tools, Materials and equipment



Notes to the trainer:

- Trainer may use small group for Identification of IoT Tools, Materials and equipment. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and ask learners to respond to the following questions:

i. What are two examples of microcontrollers or microprocessors used in IoT devices?

ii. Name one software tool used for data analytics and visualization in IoT

Step 2: Asks trainees to write provided answers on flipchart/paper.

Step 3: Asks trainees to present their findings to the whole class.

Step 4: Provides expert view and address any questions or concerns.

Step 5: Ask trainees to read the key reading 1.1.3 in the trainee manual.



Points to Remember

- **Software Tools and Equipment:**
  - ✓ Operating Systems (OS)
  - ✓ Middleware:
  - ✓ Device Management Platforms
  - ✓ Development Tools
  - ✓ Analytics Tools
  - ✓ Network Management
  - ✓ Security Software
- **Hardware Tools, Materials, and Equipment**
  - ✓ Microcontrollers
  - ✓ Sensors
  - ✓ Actuators
  - ✓ Communication Modules
  - ✓ Power Supplies
  - ✓ Prototyping Boards
- **Edge Devices, materials and Additional Equipment**
  - ✓ Breadboards and PCBs
  - ✓ Connectors and Wires
  - ✓ Multimeter, Soldering Iron and Screwdrivers and Pliers
  - ✓ Enclosures and Mounting Kits



Theoretical Activity 1.1.4: Identification of IoT Technology Challenges



Notes to the trainer:

- Trainer may use small group for Identification of IoT Technology Challenges. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions

i. What is one key challenge related to IoT security?
ii. Name one environmental impact challenge associated with IoT devices
iii. Compare IoT security from Data privacy

Step 2: Asks trainees to write provided answers on flipchart/paper.

Step 3: Asks trainees to present their findings to the whole class.

Step 4: Provides expert view and Address any questions or concerns.

Step 5: Ask trainees to read the key reading 1.1.4 in the trainee manual.

Points to Remember

- **Identification of IoT Technology Challenges**
  - ✓ **IoT Security:** Vulnerable to cyberattacks due to weak authentication, inadequate encryption, and lack of regular security updates
  - ✓ **Data Privacy:** Concerns over the collection, sharing, and potential
  - ✓ **IoT Devices Safety:** Ensuring safety and reliability, particularly in critical applications like healthcare and automotive
  - ✓ **Interoperability**: Challenges due to lack of standardization and compatibility among various IoT devices and platforms, leading to integration issues.
  - ✓ **Environmental Impact:** Significant environmental footprint from energy consumption, electronic waste, and carbon emissions

Application of learning 1.1.

You are working as a technology consultant for a large agricultural company. The company is looking to implement IoT solutions to optimize their farming operations, reduce resource waste, and increase crop yield. As an IoT consultor advise them the components they have to use while designing and implementing an IoT-based agriculture system that monitors soil conditions, weather, and crop health, and automates irrigation and pest control

Checklist for application

SN	Criteria	Indicators	Yes	No
1	IoT components are properly identified	Things are Identified		
		Internet are Identified		
		Connectivity are Identified		
2	IoT Tools, Materials and equipment are correctly described	Software and Hardware Tools are described		
		Software and Hardware Materials are described		
		Software and Hardware equipment are described		

Indicative content 1.2: Identification of IoT network technologies

Duration: 4 hrs

Theoretical Activity 1.2.1: Describe Low Range/Low Power

Notes to the trainer:

- Trainer may use small group for describing Low Range/Low Power The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. Which IoT technology is ideal for home automation and has mesh networking capabilities?

ii. What is the primary use of RFID technology in IoT applications?

iii. Which IoT technology is suitable for wearables and health monitors due to its energy efficiency and short to medium range?

Step 2: Asks any learner to write their findings on reserved place.

Step 3: Asks learners to discuss the provided answer and choose correct answers.

Step 4: Provides expert view and clarifies ideas by using didactic materials.

Step 5: Address any questions or concerns.

Step 6: Ask trainees to read the key reading 1.2.1 in the trainee manual

Points to Remember

Low Range/Low Power IoT Networks are specialized wireless communication networks designed for Internet of Things (IoT) devices that require minimal power consumption and have a limited communication range. These networks are essential for enabling efficient and sustainable connectivity for a variety of IoT applications, particularly those involving battery-powered or energy-constrained devices. Examples of Low Range/Low Power IoT networks: Zigbee, Wi-Fi 6, Bluetooth/LE, NFC, RFID and Ethernet:.

Theoretical Activity 1.2.2: Describe Large Range/Low Power

Notes to the trainer:

- Trainer may use small group for describing Large Range/Low Power. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.
- Avail the image/picture show large range IoT network technologies application

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. What is the large range/Low power IoT networks?
ii. Differentiate large range from low range
iii. Enumerate Large range IoT network Technologies
iv. What does LPWAN stand for, and what is its primary benefit?

Step 2: Asks any learner to write their findings on reserved place.

Step 3: Asks learners to discuss the provided answer and choose correct answers.

Step 4: Provides expert view and clarifies ideas by using didactic materials.

Step 5: Address any questions or concerns.

Step 6: Ask trainees to read the key reading 1.2.2 in the trainee manual.



Points to Remember

- **Large Range IoT Networks** are designed to provide long-distance communication for IoT devices. These networks are essential for connecting devices that are spread over wide areas, such as in smart cities, agriculture, and industrial IoT applications. Examples of Large Range IoT Networks: LPWAN (Low Power Wide Area Network), Cellular, LoRa (Long Range, SigFox, WiFi HaLow, 5G, LTE (Long Term Evolution and Satellite.



Application of learning 1.2.

Develop and deploy a smart agriculture system to monitor and manage crop conditions across a large farm. Explore and apply various IoT network technologies to optimize communication between sensors, controllers, and cloud-based systems. You are tasked with designing a comprehensive IoT-based crop monitoring system.

Checklist for application

SN	Criteria	Indicators	Yes	No
1	IoT Network technologies are properly identified	1.1 Low Range/Low Power is selected		
		1.2 Large Range is selected		

Indicative content 1.3: Description of IoT architecture and Protocol

Duration: 4 hrs

Theoretical Activity 1.3.1: Description of Physical Model of IoT

Notes to the trainer:

Trainer may use small group for describing Physical Model of IoT. The use of images, videos, and illustrations as didactic materials is required.

- Avail tools, materials and equipment related to this activity

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions:

i. b function of a gateway in IoT?

Step 2: Asks trainees to write provided answers on flipchart/paper.

Step 3: Asks trainees to present their findings to the whole class.

Step 4: Provides expert view and Address any questions or concerns.

Step 5: Ask trainees to read the key reading 1.3.1 in the trainee manual.

Points to Remember

- The physical model of the Internet of Things (IoT) consists of various components that work together to enable connectivity, data exchange, and interaction between devices.
  - ✓ **Device Layer**: IoT devices and sensors.
  - ✓ **Connectivity Layer**: Network interfaces and protocols.
  - ✓ **Gateway Layer**: Data aggregation and preliminary processing.

- ✓ **Cloud/Edge Layer**: Data storage and advanced analytics.
- ✓ **Application Layer**: User interfaces and application software.
- ✓ **Security Layer**: Security mechanisms throughout the model.



Theoretical Activity 1.3.2: Description Logical Model of IoT



Notes to the trainer:

- Trainer may use small group for describing Logical Model of IoT. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions

i. What does the communication layer do in the logical model of IoT?
ii. How does edge computing help IoT devices?
iii. What is the purpose of the analytics layer?

Step 2: Asks trainees to write provided answers on flipchart/paper.

Step 3: Asks trainees to present their findings to the whole class.

Step 4: Provides expert view and Address any questions or concerns.

Step 5: Ask trainees to read the key reading 1.3.2 in the trainee manual.



Points to Remember

- The logical model of the Internet of Things (IoT) defines the structure and relationships between different components and processes involved in IoT systems. It focuses on the functionality and data flow rather than the physical implementation.
  The logical model can be visualized in a sequential data flow:
  - ✓ **Data Collection**: Sensors and devices gather data.

- ✓ **Data Transmission**: Data is sent via communication protocols to the edge or cloud.
- ✓ **Edge Processing**: Local processing of data occurs at the edge.
- ✓ **Data Aggregation**: Collected data is aggregated for analysis.
- ✓ **Data Analysis**: Analytical processes derive insights from the data.
- ✓ **User Interaction**: Insights and controls are presented to users through applications.
- ✓ **Security Management**: Security measures protect data and ensure integrity throughout the process.



Theoretical Activity 1.3.3: Description of IoT protocols



Notes to the trainer:

- Trainer may use small group for describing IoT protocols. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions:

1. What is MQTT, and why is it commonly used in IoT applications?
2. How does CoAP differ from HTTP in terms of its application in IoT?
3. Name one key feature of LoRaWAN and explain its importance for IoT devices.

Step 2: Asks trainees to write provided answers on flipchart/paper.
Step 3: Asks trainees to present their findings to the whole class.
Step 4: Provides expert view and address any questions or concerns.
Step 5: Ask trainees to read the key reading 1.3.3 in the trainee manual.



Points to Remember

- IoT protocols are essential for enabling communication between IoT devices, applications, and systems. They define how data is transmitted, formatted, and processed across different networks
  - **MQTT**: A publish/subscribe messaging protocol ideal for small sensors. It features low bandwidth usage and reliable message delivery with QoS support.
  - **CoAP**: Designed for constrained devices, this web transfer protocol uses a RESTful architecture, supporting standard HTTP methods over UDP.
  - **HTTP/HTTPS**: An application layer protocol for web services, known for its security (HTTPS) but not optimized for low-power devices.
  - **AMQP**: A message-oriented middleware protocol that supports reliable business messaging and communication in IoT.
  - **DDS**: A middleware protocol for real-time data exchange in industrial applications, utilizing a publish/subscribe model.
  - **SigFox**: An LPWAN protocol for low-bandwidth IoT devices, ideal for applications like asset tracking and environmental monitoring, featuring ultra-narrowband technology.



Application of learning 1.3.

Develop a smart home energy management system that monitors and controls devices like lights, HVAC systems, and appliances to reduce energy waste and optimize usage. The system will gather data from various sensors, process it, and execute commands to manage energy consumption automatically. Throughout the project, students will explore the IoT architecture and relevant protocols to ensure efficient communication and operation.

Checklist for application

SN	Criteria	Indicators	Yes	No
1	IoT Architecture and protocols are properly described	1.1 Physical model of IoT is described		
		1.2 Logical model of IoT is described		
		1.3 IoT Protocols are described		

Indicative content 1.4: Calibration of Tools, Equipment and Materials

Duration: 4 hrs

Theoretical Activity 1.4.1: Description of calibration

Notes to the trainer:

- Trainer may use small group for describing of calibration. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions:

i. What is the purpose of calibration in maintaining measurement instruments?
ii. Why is calibration important for quality control?

Step 2: Asks trainees to write provided answers on flipchart/paper.

Step 3: Asks trainees to present their findings to the whole class.

Step 4: Provides expert view and Address any questions or concerns.

Step 5: Ask trainees to read the key reading 1.4.1 in the trainee manual.

Points to Remember

- Calibration is the process of adjusting and verifying the accuracy of measurement instruments by comparing their outputs against a known standard or reference. This ensures that the instruments provide precise and accurate measurements within specified limits**.**
- The primary purpose of calibration is to maintain the accuracy and reliability of measurement instruments.

Practical Activity 1.4.2: Calibration of tools, equipment and materials

Notes to the trainer

- The trainer may use individual work based for performing calibration techniques and procedures.
- Avail practical exercises on Calibration techniques and procedures

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask learners to read the given task:
An HVAC (heating, ventilation, and air conditioning) system in a residential community is monitored and controlled by means of a temperature sensor that is part of the smart home systems. To guarantee that these sensors produce accurate temperature readings and maintain the best possible levels of comfort and energy efficiency in the homes, regular calibration is necessary. You are asked to calibrate the temperature sensor in your capacity as an IoT technician installer who is in charge of maintaining these smart home devices.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate step by step how to calibrate tools, material and equipment. While demonstrating, explain the calibration techniques and procedures.

Step 4: Give other tasks related to the activity and asks learners to calibrate.

Step 5: Verify whether Calibration have well done according to the finding answers.

Step 6: Ask trainees to read key reading 1.4.2

Step 7: Ask trainees to perform the task provided in application of learning 1.4

Points to Remember

- **Calibration Process**:
  - ✓ **Step 1: Set the Temperature Bath**: Set the temperature bath to a known reference temperature (e.g., 0°C, 25°C, and 50°C).Allow the bath to stabilize at each temperature point.
  - ✓ **Step 2: Place Sensors in Bath**: Immerse both the temperature sensor and the reference thermometer in the bath.

- ✓ **Step 3: Record Readings**: After allowing sufficient time for the sensors to stabilize, record the readings from both the temperature sensor and the reference thermometer. Repeat this process for each temperature set point (e.g., 0°C, 25°C, 50°C).
- ✓ **Step 4: Compare and Adjust**: Compare the readings from the temperature sensor with the reference thermometer. If discrepancies are found, adjust the temperature sensor according to the manufacturer's instructions or calibration software



Application of learning 1.4.

An HVAC (heating, ventilation, and air conditioning) system in a residential community is monitored and controlled by means of a temperature sensor that is part of the smart home systems. To guarantee that these sensors produce accurate temperature readings, you are asked to calibrate the temperature sensor for this smart home.

Checklist for application

SN	Criteria	Indicators	Yes	No
1	Equipment's are well prepared	1.1 Temperature sensors are selected		
		1.2 Temperature sensors are cleaned		
		1.3 Temperature sensors are mounted		
2	Tools are well prepared	2.1 Computer is well selected		
		2.2 Software tools are selected and installed		
		2.3 Connection is well performed		
3	Calibration is accurately executed	3.1 Temperature room is Set		
		3.2 Temperature Sensors in are placed		

		3.3 Have record all readings accurately and document any deviations		
4	Calibration test is properly checked	4.1 Final checking at room temperature is performed to ensure the temperature sensor remains accurate after calibration		
		4.2 Temperature sensor is Sealed or marked to indicate the calibrated		

Indicative content 1.5: Interpretation of Manual/Datasheet

Duration: 4 hrs

Practical Activity 1.5.1: Interpretation of Manual/Datasheet

Notes to the trainer

- The trainer may use individual work based for Interpretation of Manual/Datasheet
- Avail practical exercises on Interpretation of Manual/Datasheet

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask learners to perform the task described below:
You are working on an IoT project to monitor room temperature using a DHT22 temperature and humidity sensor. Before integrating the sensor into your system, you need to interpret its datasheet to ensure proper usage and functionality

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate step by step Interpretation of Manual/Datasheet. While demonstrating, explain the Interpretation of Manual/Datasheet.

Step 4: Give other tasks related to the activity and asks learners to calibrate, monitor the procedures

Step 5: Verify whether Interpretation of Manual/Datasheet have well done according to the finding answers.

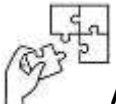
Step 6: Ask trainees to read key reading 1.5.1

Step 7: Ask trainees to perform the task provided in application of learning 1.5



Points to Remember

- **Steps of Interpretation of Manual/Datasheet**
  - ✓ **Read the Overview:** You find that the DHT22 is designed for measuring temperature and humidity accurately, making it suitable for your monitoring project.
  - ✓ **Check Electrical Specifications:** The datasheet specifies an operating voltage of 3.3V to 6V. You confirm that your power supply provides 5V, which is compatible.
  - ✓ **Review Performance Characteristics:** The measurement range for temperature is -40°C to +80°C, and the accuracy is ±0.5°C. This meets your requirements for monitoring indoor temperatures.
  - ✓ **Examine Communication Protocols:** The DHT22 uses a single-wire digital signal for communication. You check your microcontroller's capabilities and confirm it can support this protocol.
  - ✓ **Look at Mechanical Characteristics:** The datasheet includes dimensions and pin configuration, helping you plan the physical layout of your sensor in the project.
  - ✓ **Analyze Environmental Specifications:** The sensor operates within a temperature range of -40°C to +80°C and a humidity range of 0% to 100%. You ensure these conditions are suitable for your intended use.
  - ✓ **Follow Calibration Instructions:** The datasheet states the sensor comes factory calibrated, so you won't need to perform additional calibration, simplifying your setup.
  - ✓ **Study the Application Circuit:** The provided schematic shows how to connect the sensor to your microcontroller, guiding you in the wiring process.



Application of learning 1.5.

Gasabo Plaza is commercial building which their owner want install HVAC for temperature and Humidity conditions. As IoT technician you are asked to interpret the datasheet effectively to ensure that the selected device will meet your IoT project requirements and integrate smoothly into your system**.**

Checklist /Solution for application

SN	Criteria	Indicators	Yes	No
1	The temperature sensor is well selected	1.1 The DHT22 measures both temperature and humidity is well selected.		
		1.2 additional features (e.g., digital output) are correctly identified		
2	Electrical Characteristics are properly indicated	2.1 Operating Voltage is well chosen		
		2.2 Power Consumption in Different Modes is assessed		
		2.3 Wake-up Time is checked		
3	Performance Specifications are correctly described	3.1The temperature range are checked		
		3.2 The humidity range are checked		
4	Physical Characteristics are correctly described	4.1 The DHT22's dimensions fit are verified		
		4.2 Pinout and Connection Interface (VCC, GND, DATA) are checked		
		4.3 The operating environment is selected		
5	Communication Protocol are correctly described	5.1 Data Interface is verified		
		5.2 Data Rate and Timing is Reviewed		



Learning outcome 1 end assessment

Written assessment

1. **Which of the following is NOT a benefit of IoT?**

    a) Enhanced Efficiency

    b) Improved Decision Making

    c) Increased Manual Labor

    d) Cost Savings

    Answer c) Increased Manual Labor

2. **Which protocol is commonly used for low-power IoT devices to communicate?**

    a) HTTP

    b) MQTT

    c) FTP

    d) SMTP

    Answer: b) MQTT

3. **What is the primary purpose of calibration in IoT?**

    a) To improve the physical appearance of devices

    b) To ensure the accuracy and reliability of measurements

    c) To increase the speed of data transmission

    d) To reduce the size of IoT devices

    Answer: b) To ensure the accuracy and reliability of measurements

4. **Match the IoT components with their descriptions:**

Column A: IoT System Components	**Column B: Functions**
1. Sensors	a) Execute actions based on data received
2. Actuators	b) Collect data from the environment
3. Connectivity Modules	c) Process data received from sensors

4. Microcontrollers/Microprocessors	d)Enable communication between devices

Answer:

1- b) Collect data from the environment
2 - a) Execute actions based on data received
3 - d) Enable communication between devices
4 - c) Process data received from sensor

5. **Match the following table**

Column A: IoT System Components	**Column B: Functions**
1.Sensors and Actuators	a. Facilitates efficient energy usage in devices
2.Communication Protocols	b. Defines how devices communicate in the network
3.Data Storage	c. Stores and manages data collected from devices
4.Power Management	d. Collect and transmit data from the physical world

Answers:

1.d - Collect and transmit data from the physical world
2.b - Defines how devices communicate in the network
3.c - Stores and manages data collected from devices
4.a - Facilitates efficient energy usage in devices

6. Answer by True or False in statement below
   a. IoT devices always guarantee data privacy. True / False

   Answer: False
   b. Zigbee and Bluetooth are examples of large range IoT network technologies. True / False

   Answer: False
   c. IoT devices can help improve decision-making by providing real-time data. True / False

   Answer: True
   d. NFC and RFID are examples of large range IoT network technologies. True / False

   Answer: False

Practical assessment

A&B is company in charge of Preparing IoT system installation. They need to prepare smart lights in a residential setting. Your focus is on the meticulous steps leading up to the actual installation of the smart lights. As a skilled technician, you've been assigned to prepare for the installation of an IoT smart lighting system in a residential. Your tasks involve the systematic preparation, including the identification of IoT network technology, description of IoT architecture and protocols, and the calibration of tools, equipment, and materials.

The duration of the work is 3 hours.

Resources

Tools	Plier, screwdriver and Web browser software
Equipment	IoT-enabled Smart Light Bulbs, Home Automation Hub (e.g., SmartThings Hub), Mobile Devices (Smartphones), Wi-Fi Network and Existing Lighting Fixtures
Material/Consumable	Internet bundles, power extension, electricity

SN	Criteria	Indicators	Yes	No
1.	Safety and security measures are effectively applied	1.1 PPEs are worn and are effectively applied		
		1.2 Safety equipment are used		
		1.3 Health and security measures are applied		
2	IoT Tools, Materials and equipment are properly identified	2.1 Equipment are selected		
		2.2 Materials are selected		
		2.3 Software and hardware Tools are selected		
3	IoT components are correctly identified	3.1 Things are selected		
		3.2 Internet is available		
		3.3 Connectivity is well selected		
4		4.1 IoT Security is applied		

	IoT Technology Challenges are properly identified	4.2 Data Privacy is applied		
		4.3 IoT Devices Safety is applied		
		4.4 Interoperability is applied		
		4.5 Environmental Impact is checked		
5	IoT Network Technologies are correctly identified	5.1 IoT Network Technologies are identified		
6	IoT Architecture and protocols are properly described	6.1 Physical model of IoT is designed		
		6.2 Logical model of IoT is drawn		
		6.3 IoT Protocols is selected		
7	Tools, equipment and materials are correctly calibrated	7.1 Tools are calibrated		
		7.2 Equipment is calibrated		
		7.3 Materials are calibrated		
8	Manual/Datasheet are correctly interpreted	8.1 Manual is interpreted		
		8.2 Datasheet is interpreted		

END



Further information to the trainer

"Architecting the Internet of Things: State of the Art" *By Dieter Uckelmann, Mark Harrison, Florian Michahelles*

"Internet of Things: Principles and Paradigms" *Edited by Rajkumar Buyya, Amir V

"Internet of Things: Technologies and Applications for a New Age of Intelligence" *By Michael Miller*

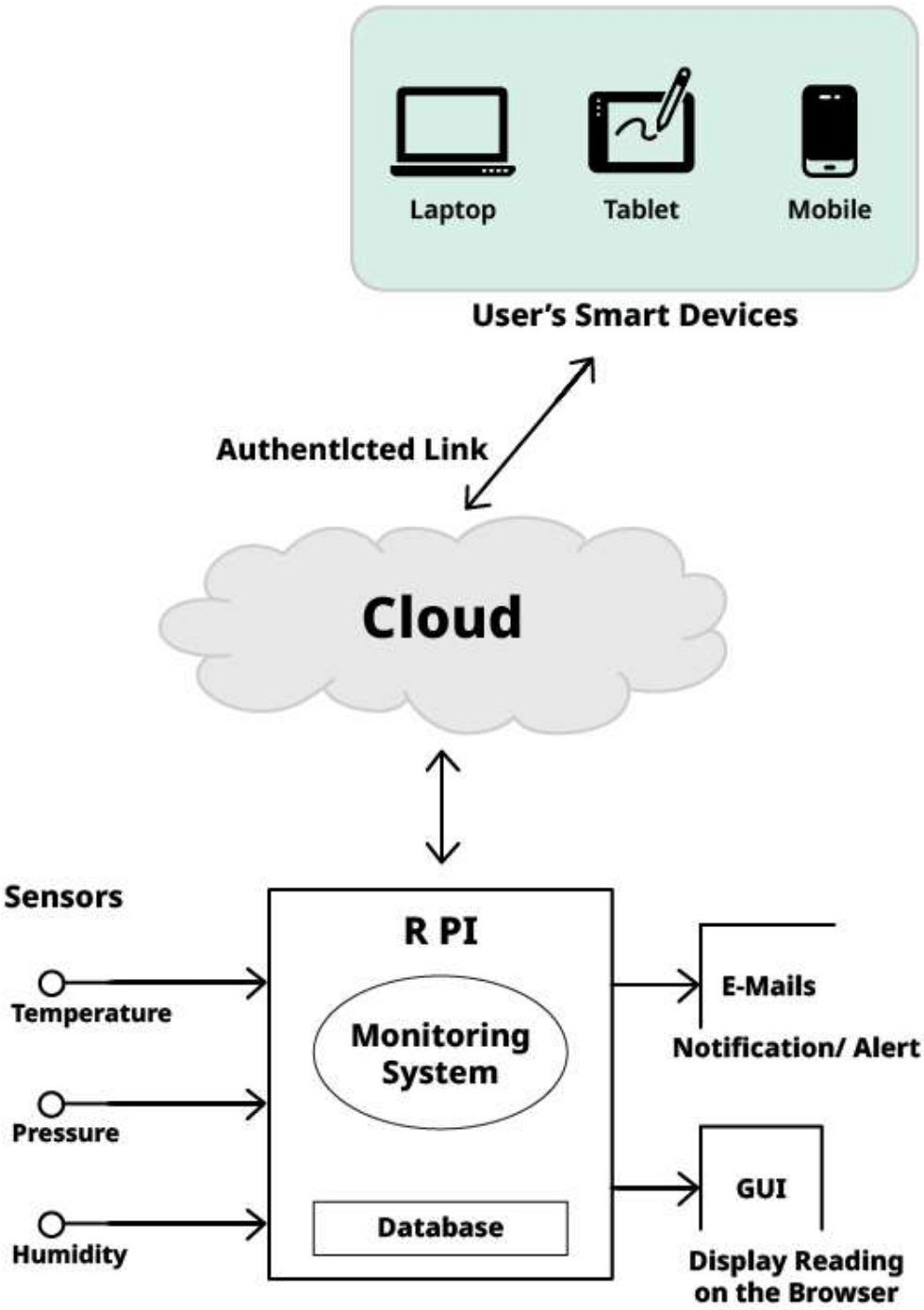
"Introduction to Measurement and Instrumentation" *By Robert G. Seipp*

"IoT Networks and Communications" By M. A. G. Nair, S. V. M. Chavan.

"Measurement and Calibration: A Practical Approach" *By David W. Ziegler*

"Precision Instrumentation and Calibration: Concepts, Techniques, and Applications" *By Gary R. Ruddell*

"The Internet of Things: A Look at Real-World Use Cases and the Future of IoT" *By Richard J. K. Liu*


Laptop
Tablet
Mobile
User's Smart Devices
AuthentIcted Link
Cloud
Sensors
Temperature
Pressure
Humidity
R PI
Monitoring System
Database
E-Mails
Notification/ Alert
GUI
Display Reading on the Browser

Indicative contents
2.1 Performance of workplace Setup
2.2 Identification of IoT deployment levels
2.3 Assembling of IoT equipment
2.4. Configuration of IoT devices
2.5 Implementation of IoT system Security
2.6 Documentation of IoT installation report

Key Competencies for Learning Outcome 2: Deploy IoT Equipment

Knowledge	Skills	Attitudes
• Description of basics of IoT • Description of Communication Protocols of IoT • Identification of IoT deployment levels	• Assembling of IoT Equipment • Networking Setup and Configuration • Making Integration System • Installing of Firmware/Software • Configurating of IoT Platform • Implementing of Security • Preparing documentation of IoT Installation Report	• Having Collaboration attitudes • Being Problem-solver • Having a Perseverance • Having an Initiative



Duration: 10hrs

Learning outcome 2 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Set properly the workplace according to the installation design
2. Assemble correctly IoT equipment as per the manufacturer's guides
3. Customize appropriately the equipment settings based on the installation
4. Apply properly IoT system protection measures as per the installation manual
5. Perform correctly Functional testing according to the installation requirements
6. Report properly IoT system installation based on the installation requirements



Resources

Equipment	Tools	Materials
● Modem, ● Access Point, ● Computer, ● Microcontrollers ● Arduino Starter Kit	● Software platforms, ● Simulation ● software, ● Tape measure, ● Multi-meter, ● Screwdrivers, ● Drilling machine, ● Hammer, ● Ladder, ● Pliers	● Internet connectivity, ● Pipes, ● Cable trunks, ● Plectrical tapes, ● Screws, ● Labelling tags



Advance Preparation:

Before delivering this learning outcome, you are recommended to:

- Avail IoT devices, tools, equipment and materials
- Ensure that the installation environment is ready
- Avail resources for participants to further explore IoT system deploy equipment

Indicative content 2.1: Performance of Workplace Setup

Duration: 1 hr

Theoretical Activity 2.1.1: Study the workplace environment

Notes to the trainer:

- Trainer may use small group for studying the workplace environment. The use of images, videos, and illustrations as didactic materials is required.
- Avail the specific areas where IoT technologies can potentially be implemented
- Evaluate the current IoT infrastructure and devices deployed in our workplace.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. How does your commute distance impact your overall job satisfaction and punctuality?

ii. Are there any physical barriers in your workspace that make it difficult for you to communicate or move around?

iii. Do you feel that the current lighting and HVAC systems in your workplace provide a comfortable and productive environment?

Step 2: Asks trainees to write provided answers on flipchart/paper.

Step 3: Asks trainees to present their findings to the whole class.

Step 4: Provides expert view and address any questions or concerns.

Step 5: Ask trainees to read the key reading 2.1.1 in the trainee manual.

Points to Remember

- ✓ **Distance:**
  - Evaluate the average distance employees travel to work, as it affects punctuality, stress, and job satisfaction.
  - Ensure optimal spacing between workstations, departments, and facilities (like restrooms and break rooms) to promote efficiency and collaboration.
  - Confirm that emergency exits are easily accessible from all areas.
- ✓ **Physical Obstacles**

- Identify and manage physical barriers (walls, partitions, furniture) that might impede movement or communication.
- Ensure the workplace is accessible to everyone, including those with disabilities, with features like ramps, elevators, and wide corridors.
- Check that workstations are ergonomically designed to prevent physical strain, considering desk height, chair type, and monitor placement.

✓ **Existing Systems (Source of Power)**

- Ensure there are sufficient power outlets in convenient locations to avoid overloading circuits.
- Assess the lighting to ensure it is adequate for tasks without causing glare or eye strain, and maximize natural light where possible.
- Evaluate the HVAC system to maintain a comfortable temperature and good air quality.
- Confirm that the IT infrastructure supports the organization's technological needs reliably.
- Verify the presence and functionality of safety systems, including fire alarms, sprinklers, and emergency lighting.



Practical Activity 2.1.2: Selection of requirements



Notes to the trainer

- The Trainer may use individual based facilitation for Selection of requirements. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform to the following task:
I&B company ltd need to select Tools, Materials, and Equipment for an IoT Smart Home Automation System. The system will encompass: Lighting control, Climate control (heating and cooling), Security monitoring (cameras, door/window sensors, smart locks), Appliance control (smart plugs, connected appliances) and Energy monitoring and management. As an IoT system installer you are assigned to specify selection of requirements for this system.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to select the requirements of the given tasks. While demonstrating, explain Step-by-Step Instructions procedures

Step 4: Give other tasks related to the activity and Asks learners to select, monitor the procedures.

Step 5: Verify whether selections have well done according to the finding answers.

Step 6: Ask trainees to read key reading 2.1.2

Step 7: Ask trainees to perform the task provided in application of learning 2.1



Points to Remember

- **Steps for selecting requirements**
  - ✓ Define Project Requirements
  - ✓ Identify Key Components
  - ✓ Determine Connectivity Options
  - ✓ Plan Power Supply
  - ✓ Choose Software Tools
  - ✓ Source Materials and Equipment



Theoretical Activity 2.1.3: Mapping the workplace



Notes to the trainer:

- Trainer may use small group for mapping the workplace. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.
- Avail layout plan of workplace



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. What is mapping workplace?

ii. Why is necessary to map the workplace?

Step 2: Asks any learner to write their findings on reserved place.

Step 3: Asks learners to discuss the provided answer and choose correct answers.

Step 4: Provides expert view and clarifies ideas by using didactic materials.

Step 5: Address any questions or concerns.

Step 6: Ask trainees to read the key reading 2.1.3 in the trainee manual.

Points to Remember

- **Workplace mapping** is one of the most effective ways of examining the strength and weaknesses of your branch. Mapping is the process of obtaining accurate and relevant information about the workforce in a systematic way. If done properly it can form the basis of most successful union activity, including recruiting, communications, developing activists and campaigning around issues

Practical Activity 2.1.4: Setting up network connectivity

Notes to the trainer

- The trainer may use individual facilitation based; you are supposed to Setup network connectivity;
- Avail examples of different of tools, materials and equipment.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask learners to perform the task described below:
In Small Office, they want to setup network connectivity for IoT devices. You are tasked with setting up network connectivity for a smart office environment, integrating IoT devices to enhance efficiency and connectivity among multiple devices

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate step by step how to set network connectivity. While demonstrating, explain the procedures.

Step 4: Give other tasks related to the activity and Asks learners to setup, monitor the procedures.

Step 5: Verify whether setting up has well done according to the finding answers.

Step 6: Ask trainees to read key reading 2.1.4

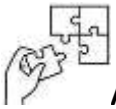
Step 7: Ask trainees to perform the task provided in application of learning 2.1.



Points to Remember

- **Steps for Setting up network connectivity**

✓ Begin by configuring the router to establish a smart network backbone for the office.
✓ Integrate IoT devices to automate office tasks and enhance productivity.
✓ Establish both wired and wireless connections to support a variety of office devices.
✓ Implement robust security measures to protect the smart office network and IoT ecosystem.
✓ Conduct comprehensive testing to ensure the smart office IoT network meets performance expectations.



Application of learning 2.1.

KGL company want to set up the new office space, it has the role of studying the environment to ensure optimal setup for efficiency and productivity. Choose the right tools, equipment, and materials necessary for setting up a functional and efficient workplace. Establish a reliable and secure network infrastructure to perform the workplace setup for the new office operations.

Checklist for application

SN	Criteria	Indicators	Yes	No
1	Study the workplace environment is well performed	1.1 Distance is determined		
		1.2 Physical obstacles is checked		
		1.3 Source of power is availed		
2	Requirements are correctly selected	2.1 Tools are selected		
		2.2 Equipment are selected		
		2.3 Materials are selected		
3	Workplace is properly mapped	3.1 Workplace is mapped		
4	Network connectivity is well set up	4.1 Network connectivity is set up		

Indicative content 2.2: Identification of IoT Deployment Levels

Duration: 2 hrs

Theoretical Activity 2.2.1: Description of IoT deployment levels

Notes to the trainer:

- Trainer may use small group for describing IoT deployment levels. The use of images, videos, and illustrations as didactic materials is required. Avail tools, materials and equipment related to this activity

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:
   i. Give a list of steps or information related to IoT network connectivity setups?

Step 2: Asks any learner to write their findings on reserved place.

Step 3: Asks learners to discuss the provided answer and choose correct answers.

Step 4: Provides expert view and clarifies ideas by using didactic materials.

Step 5: Address any questions or concerns.

Step 6: Ask trainees to read the key reading 2.2.1 in the trainee manual.

Points to Remember

- **Level-1: Device Connectivity:** Establishes basic connectivity for IoT devices. Devices transmit data to a centralized system. Examples: Smart home devices, basic industrial sensors.
- **Level-2: Device Management and Monitoring:** Focuses on managing and monitoring connected devices. Includes remote monitoring and basic management functionalities. Examples: Fleet management, basic home automation.
- **Level-3: Data Integration and Analytics:** Integrates data from multiple devices into a centralized platform. Applies basic analytics for insights and monitoring. Examples: Energy management, predictive maintenance.

- **Level-4: Advanced Analytics and Automation:** Utilizes advanced analytics (e.g., machine learning) for deeper insights. Enables real-time decision-making and operational optimization. Examples: Supply chain optimization, smart agriculture.
- **Level-5: Autonomous Systems:** Implements AI-driven decision-making and autonomous operations. Systems operate with minimal human intervention. Examples: Autonomous vehicles, smart cities infrastructure.
- **Level-6: Integrated Ecosystems:** Fully integrates IoT devices, systems, and platforms. Ensures interoperability and scalability across domains. Examples: Smart cities, industrial IoT platforms.



Application of learning 2.2

QZ company is a large-scale farm which wants to implement a smart irrigation system to optimize water usage, increase crop yield, and reduce labor costs. The farm is divided into multiple zones, each growing different crops with varying water needs. The goal is to automate the irrigation process based on real-time data from the field. Describe the IoT deployment levels that will be applied.

Checklist for application

SN	Criteria	Indicators	Yes	No
1	IoT deployment levels are properly identified	1.1 Device Level (Level 1) is identified		
		1.2 Network Level (Level 2) is identified		
		1.3 Edge Level (Level 3) is identified		
		1.4 Cloud Level (Level 4) is identified		
		1.5 Application Level (Level 5) is identified		
		1.6 Business Level (Level 6) is identified		



Indicative content 2.3: Assembling of IoT equipment

Duration: 2 hrs

Theoretical Activity 2.3.1: Manufacturer's guide interpretation

Notes to the trainer:

- Trainer may use small group for interpreting Manufacturer's guide. The use of images, videos, and illustrations as didactic materials is required.
- Avail manufacturer's guides provide crucial instructions and information for proper assembly of IoT equipment.
- Avail tools, materials and equipment related to this activity

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. Define the manufacturer's guides
ii. What is the importance of manufacturer's guides

Step 2: Asks any learner to write their findings on reserved place.
Step 3: Asks learners to discuss the provided answer and choose correct answers.
Step 4: Provides expert view and clarifies ideas by using didactic materials.
Step 5: Address any questions or concerns.
Step 6: Ask trainees to read the key reading 2.3.1 in the trainee manual.

Points to Remember

- **Definition of Manufacturer's Guides:** Manufacturer's guides are documents provided by product manufacturers that contain instructions, specifications, and recommendations for the installation, operation, and maintenance of their products.
- **Importance of Manufacturer's Guides:** Manufacturer's guides provide crucial information on how to correctly install, operate, and maintain products, ensuring optimal performance and longevity. They help users avoid errors, accidents, and damage to the product, as well as ensure compliance with safety regulations and warranty requirements



Practical Activity 2.3.2: Fixing of IoT equipment



Notes to the trainer

- The trainer may use individual based Facilitation of this activity; you are supposed to Fix IoT equipment; it is recommended to: Avail examples and exercises on Fixing of IoT equipment.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask learners to perform the task described below:
We want to mount and fix IoT equipment in a smart office environment for energy efficiency, space utilization, and employee well-being.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate step by step how to fix IoT equipment. While demonstrating, explain the calculation procedures.

Step 4: Give other tasks related to the activity and Asks learners to fix, monitor the procedures.

Step 5: Verify whether fixations have well done according to the finding answers.

Step 6: Ask trainees to read key reading 2.3.2

Step 7: Ask trainees to perform the task provided in application of learning 2.3



Points to Remember

Steps to follow when fixing IoT devices:

- ✓ Evaluate the office layout, identifying key areas for deploying IoT devices, such as workstations, meeting rooms, and common areas.
- ✓ Choose IoT devices based on the office requirements, including smart lighting for energy efficiency, occupancy sensors for workspace utilization, and environmental sensors for comfort monitoring.
- ✓ Mount smart lighting fixtures in common areas and workstations, ensuring even distribution.
- ✓ Mount occupancy sensors in strategic locations, such as above workstations and in meeting rooms.

- ✓ Install temperature and humidity sensors in areas where environmental conditions may impact employee comfort. Consider mounting sensors away from direct airflow or heat sources for accurate readings.
- ✓ Place the IoT gateway in a central location within the office to ensure effective communication with all deployed devices.
- ✓ Connect sensors to the IoT gateway using appropriate wiring and connectors. Conceal wiring where possible to maintain a clean and organized appearance.
- ✓ Configure each IoT device through the manufacturer's application or a centralized control app. set up automation rules, such as adjusting lighting based on occupancy or optimizing HVAC settings for comfort.
- ✓ Test each device individually to ensure proper functionality. Verify that sensors accurately detect occupancy, control lighting, and provide real-time environmental data.



Application of learning 2.3.

Assemble an IoT-based temperature monitoring system for a smart home that collects real-time temperature data and sends it to a central hub, which can be accessed via a smartphone app. Components Required are Microcontroller, Temperature Sensor, Breadboard and Jumper Wires, 5V USB power supply, Wi-Fi Router, IoT Platform and a Smartphone with IoT App: Blynk app or a custom-built app to monitor temperature.

Checklist for application

SN	Criteria	Indicators	Yes	No
1	Required components are correctly selected	1.1 Microcontroller of ESP8266 or ESP32 is selected		
		1.2 Temperature Sensor of DHT22 or DS18B20 is selected		
		1.3 Breadboard and Jumper Wires are avail		
		1.4 Power Supply of 5V is avail		
		1.5 Wi-Fi Router is well selected		
		1.6 IoT Platform is chosen		

		1.7 Smartphone with IoT App is availed		
2	IoT equipment are correctly assembled	2.1 Manufacturer's guide is used		
		2.2 IoT equipment is Fixed		
		2.3 All components are assembled		

Indicative content 2.4: Configuration of IoT devices

Duration: 2 hrs

Theoretical Activity 2.4.1: Selection of IoT platform

Notes to the trainer:

- Trainer may use small group for Selecting IoT platform. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. Explain the on-premise IoT platform is an Internet of Things
ii. State the advantages of on-premise IoT platform
iii. Differentiate on premise to cloud IoT platform

Step 2: Asks any learner to write their findings on reserved place.
Step 3: Asks learners to discuss the provided answer and choose correct answers.
Step 4: Provides expert view and clarifies ideas by using didactic materials.
Step 5: Address any questions or concerns.
Step 6: Ask trainees to read the key reading 2.4.1 in the trainee manual.

Points to Remember

- On-premise definition: On-premise is the IT infrastructure and software located within an organization's physical office and hosted on-site. In other words, on-premise means the software is installed on physical hardware that is owned by an organization, located in the physical premises of the organization, usually in the organization's own data centre

- The cloud refers to the software, servers, and services that run over the internet rather than locally on the computers and hardware of the organization. Cloud servers are located all around the world in different big data centres. You can access

cloud services through your web browsers, such as Google Chrome and Firefox. Some cloud service providers also have dedicated mobile apps to access cloud services



Practical Activity 2.4.2: Installation of IoT applications



Notes to the trainer

- The trainer may use individual based Facilitation of this activity; you are supposed to Fix IoT equipment; it is recommended to: Avail examples and exercises on Fixing of IoT equipment.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask learners to perform the task described below:
Download and install the Smart Home (MSmartHome) IoT software application on your smartphone, enabling you to control and monitor your smart devices.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate step by step how to install IoT application. While demonstrating, explain the installation procedures.

Step 4: Give other tasks related to the activity and Asks learners to install, monitor the procedures.

Step 5: Verify whether installation has well done according to the finding answers.

Step 6: Ask trainees to read key reading 2.4.2

Step 7: Ask trainees to perform the task provided in application of learning 2.4



Points to Remember

- **IoT Device provisioning**
  IoT device provisioning is the process of on boarding and configuring IoT devices so that they can securely connect to a network, communicate with other devices or services, and perform their intended functions. Proper device provisioning is critical for the successful deployment and operation of IoT solutions
- **IoT Management (Future Ready device management eg: remote configuration, Unified Device Management)**

- ✓ **Prepare Your Smartphone:** Ensure that your smartphone is connected to a stable Wi-Fi network or has an active mobile data connection. Make sure your smartphone has sufficient storage space for the application.
- ✓ **Identify the Smart Home IoT Application:** Determine the specific Smart Home IoT application that corresponds to your smart devices. This information is usually provided by the manufacturer or can be found on their official website.
- ✓ **Access App Store (iOS) or Google Play Store (Android):** Open the App Store if you are using an iPhone (iOS) or the Google Play Store if you are using an Android device.
- ✓ **Search for the Smart Home IoT Application**: In the search bar, type the name of the Smart Home IoT application you identified in step 2.
- ✓ **Select the App:** From the search results, select the correct Smart Home IoT application. Ensure that it is developed by the official manufacturer or a reputable developer.
- ✓ **Check Compatibility:** Verify that the application is compatible with your smartphone's operating system version (iOS or Android). This information is typically visible on the app's store page.
- ✓ **Read Reviews and Ratings:** Before downloading, check user reviews and ratings to get an idea of the application's performance and reliability.
- ✓ **Download the Application**: Tap the "Download" or "Install" button on the app's store page. The application will begin to download and install on your smartphone.
- ✓ **Wait for Installation:** Allow the application to complete the download and installation process. This may take a few minutes, depending on your internet connection speed.
- ✓ **Open the Application:** Once the installation is complete, tap the "Open" button to launch the Smart Home IoT application.
- ✓ **Create an Account (if required):** Some Smart Home IoT applications may require you to create an account. Follow the on-screen instructions to set up an account if needed.
- ✓ **Log In (if applicable):** If you created an account, log in with your credentials. If not, proceed with the provided options, such as connecting to your smart home devices
- ✓ **Add Smart Devices:** Follow the application's instructions to add and connect your smart devices to the app. This often involves scanning QR codes, pressing buttons on devices, or entering device-specific information.
- ✓ **Configure Settings:** Customize settings within the Smart Home IoT application according to your preferences. This may include naming devices, setting automation rules, or adjusting security settings.

- ✓ **Start Controlling and Monitoring:** Once your smart devices are added and configured, you can start using the Smart Home IoT application to control and monitor them from your smartphone



Practical Activity 2.4.3: Functional testing



Notes to the trainer

- The trainer may use individual based Facilitation of this activity; you are supposed to apply Functional testing; it is recommended to: Avail examples and exercises on Functional testing



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask learners to perform the task described below:

You are an IoT technician; you are asked to perform functional testing on an IoT system designed for real-time data transmission in a smart manufacturing environment. The goal of this testing is to assess the system's network performance, focusing on speed, latency, jitter, and quality of service (QoS) to ensure it meets the requirements of the manufacturing process. To perform functional testing to evaluate the IoT network's speed, latency, jitter, and QoS parameters to determine if it can support real-time data transmission in a smart manufacturing environment.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to test the given tasks. While demonstrating, explain the testing procedures.

Step 4: Give other tasks related to the activity and Asks learners to test, monitor the procedures.

Step 5: Verify whether installation has well done according to the finding answers.

Step 6: Ask trainees to read key reading 2.4.3

Step 7: Ask trainees to perform the task provided in application of learning 2.4



Points to Remember

IoT Functional testing: Functional testing in the context of IoT Cloud Applications includes the testing of the scenarios, such as user actions on the applications, data and events coming from the devices, and user actions

- **Speed:** Speed refers to the maximum rate you can transmit data, typically measured as megabits per second (Mbps). Bandwidth refers to the maximum amount of data your connection can handle at any moment, also measured as Mbps (and increasingly Gbps, for gigabyte connections).
- **Latency:** If you are unfamiliar with latency, it is the delay of time between the initiation of a command or input from one side to the reception from the other. It is usually measured in milliseconds (ms).
- **Jitter:** Jitter is when there is a time delay in the sending of these data packets over your network connection. This is often caused by network congestion, and sometimes route changes.
- **QoS:** Quality of service refers to the effectiveness of a network to provide more suitable or appropriate service to the application**.**



Application of learning 2.4.

You are tasked with configuring a smart HVAC (Heating, Ventilation, and Air Conditioning) system for a residential community. The system includes IoT-enabled thermostats, temperature sensors, and a central control unit that communicates with the cloud for data analytics and remote management. The objective is to install the IoT applications on the devices, configure them, and perform functional testing to ensure proper operation.

Checklist /Solution for application

SN	Criteria	Indicators	Yes	No
1	IoT Applications are properly installed	1.1 Devices are powered up		
		1.2 Each device is Configured on Network		
		1.3 Smart Thermostat App is installed		
		1.4 Sensor Configuration App is installed		
		1.5 Central Control Unit software is Installed		

		1.6 Device are registered and Cloud integrated		
		1.7 Software is updated		
2	Functional testing is correctly performed	2.1 Thermostat and temperature sensor are properly connected to the network		
		2.2 Sensor are calibrated		
		2.3 The smart thermostats are tested by different settings		

Indicative content 2.5: Implementation of IoT system Security

Duration: 2 hrs

Theoretical Activity 2.5.1: Setup the security measures

Notes to the trainer:

- Trainer may use small group for Setting up the security measures. The use of images, videos, and illustrations as didactic materials is required. Avail tools, materials and equipment related to this activity

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following question
Differentiate Physical from Logical Setup the security measures
Step 2: Asks any learner to write their findings on reserved place.
Step 3: Asks learners to discuss the provided answer and choose correct answers.
Step 4: Provides expert view and clarifies ideas by using didactic materials.
Step 5: Address any questions or concerns.
Step 6: Ask trainees to read the key reading 2.5.1 in the trainee manual.

Points to Remember

Setup the security measures

- **Physical:** Physical security involves the use of multiple layers of interdependent systems that can include CCTV surveillance, security guards, protective barriers, locks, access control, perimeter intrusion detection, deterrent systems, fire protection, and other systems designed to protect persons and property.
- **Logical:** Logical access control is the means by which organizations implement security constraints over proprietary data and broader digital environments. Various logical access control measures at the individual and team levels make up a broader logical security policy at the organizational level. By regulating access in this way, a business can prevent any single user from having too much control over an entire enterprise's digital assets.



Practical Activity 2.5.2: Perform IoT device firmware Update



Notes to the trainer

- The trainer may use individual based Facilitation of this activity; you are supposed to Perform IoT device firmware Update; it is recommended to: Avail examples and exercises on Functional testing



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask learners to perform the task described below:
You are the IoT administrator for a residential complex that uses smart thermostats to control heating and cooling in individual units. It has been determined that a firmware update is necessary to enhance security and add new features to the smart thermostats. As an IoT administrator you are tasked to perform versioning and control IoT firmware update for the smart thermostats

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how perform IoT firmware update the given tasks. While demonstrating, explain the testing procedures

Step 4: Give other tasks related to the activity and Asks learners perform IoT firmware update, monitor the procedures

Step 5: Verify whether the update has well done according to the finding answers.

Step 6: Ask trainees to read key reading 2.5.2

Step 7: Ask trainees to perform the task provided in application of learning 2.5

Points to Remember

- **Performing IoT Device Firmware Update**

  Steps:

  ✓ Check for Updates: Regularly check for firmware updates from the device manufacturer.

  ✓ Download Update: Securely download the firmware update package.

  ✓ Backup Configuration: Backup the current configuration and settings of the device.

  ✓ Authenticate Update: Verify the authenticity of the firmware using digital signatures or checksums.

  ✓ Deploy Update: Upload the firmware to the device, either through a management console or over-the-air (OTA) updates.

  ✓ Validate Update: Confirm the update was successful by checking the device status and functionality.

  ✓ Restore Configuration: If necessary, restore the previous configuration and settings.

  ✓ Monitor Device: Ensure the device operates correctly post-update and monitor for any issues.

Application of learning 2.5.

A smart warehouse relies on a network of IoT devices, including RFID tags for inventory management. The warehouse management system (WMS) has received notifications from several device manufacturers indicating that new firmware updates are available. Your task is to perform a firmware update on IoT devices within a smart warehouse environment to enhance security, fix bugs, and introduce new features.

Checklist for application

SN	Criteria	Indicators	Yes	No
1	Update are well prepared	1.1 Devices for Update are identified		
		1.2 Configurations and Data are Backed up		
		1.3 Test Firmware in a Controlled Environment		
		1.4 Results are tested		

2	Firmware Versioning and Control are properly performed	2.1 The update process is initiated		
		2.2 Versioning standards are established		
		2.3 Firmware Control Measures are taken		
3	Firmware Update are properly executed	3.1 Firmware versioning is performed		
		3.2 Firmware control is performed		
		3.3 Regular Firmware are checked		

Indicative content 2.6: Documentation of IoT installation report

Duration: 1 hr

Theoretical Activity 2.6.1: Data and facts recording

Notes to the trainer:

- Trainer may use small group for Setting up the security measures. The use of images, videos, and illustrations as didactic materials is required. Avail tools, materials and equipment related to this activity

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following question:

i. What information should be recorded about the IoT devices installed, and why is it important?

ii. What details should be captured regarding network connectivity during an IoT installation?

Step 2: Asks any learner to write their findings on reserved place.

Step 3: Asks learners to discuss the provided answer and choose correct answers.

Step 4: Provides expert view and clarifies ideas by using didactic materials.

Step 5: Address any questions or concerns.

Step 6: Ask trainees to read the key reading 2.6.1 in the trainee manual.

Points to Remember

- Project Overview
  - ✓ Project Information
  - ✓ Installation Purpose
- Device Details
  - ✓ Device Specifications
  - ✓ Installation Location
  - ✓ Mounting and Power Details
  - ✓ Network Connectivity
  - ✓ Initial Configuration

- ✓ Integration
- Testing and Validation
  - ✓ Testing Procedures
  - ✓ Validation Checks
- Issues and Resolutions
  - ✓ Issues Encountered
  - ✓ Recommendations
  - ✓ Documentation and Appendices
  - ✓ Installation Diagrams and Photos
  - ✓ Device Manuals and Specifications
  - ✓ Test Results and Validation Documents



Practical Activity 2.6.2: Report generation



Notes to the trainer

- The trainer may use individual based Facilitation of this activity; you are supposed to generate a report; it is recommended to: Avail examples and exercises on data facts



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask learners to perform the task described below:
You are asked to generate a report for any IoT system
Step 2: Explain the task and provide clear work instruction (Task, Time allocated)
Step 3: Demonstrate how to generate a report, the given tasks. While demonstrating, explain the reporting procedures.
Step 4: Give other tasks related to the activity and Asks learners to generate a report, monitor the procedures.
Step 5: Verify whether report has well done according to the finding answers.
Step 6: Ask trainees to read key reading 2.6.2
Step 7: Ask trainees to perform the task provided in application of learning 2.6

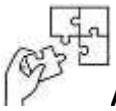


Points to Remember

- IoT Installation Report

- ✓ Project Overview
- ✓ Objectives
- ✓ System Description

- IoT Devices Installed
- Connectivity
- Software Used:
  - ✓ Installation Details
  - ✓ Data Collection and Analysis
  - ✓ Maintenance and Support
  - ✓ Conclusions and Recommendations
  - ✓ Appendices



Application of learning 2.6.

A large farm uses an IoT-based smart agriculture system to monitor soil moisture, weather conditions, crop health, and irrigation levels across different zones. The farm manager needs to generate a detailed report at the end of the growing season to analyze the performance of the crops, assess the efficiency of water usage, and make data-driven decisions for the next season. You as IoT technician generate the report for this large farm.

Checklist for application

SN	Criteria	Indicators	Yes	No
1	Project is correctly overviewed	Project Name is indicated		
		Client/Organization is indicated		
		Location is indicated		
		Installation Date is indicated		
		Report Prepared By is indicated		
		Contact Information is indicated		
2	Objectives are well indicated	Primary Goal is indicated		
		Specific Objectives are indicated		

3	System is correctly described	IoT Devices are described		
		Connectivity is described		
		Used software are described		
	Installation Details are correctly indicated	Installation Team is indicated		
		Installation Process is indicated		
		Challenges Faced are indicated		
	Data are properly collected and analyzed	Data Points Collection is indicated		
		Data Storage type indicated		
		Data Analysis Results type indicated		
	Conclusions and Recommendations are correctly indicated	Conclusions are indicated		
		Recommendations are indicated		



Learning outcome 2 end assessment

Written assessment

Choose the letter corresponding to the correct answer:

1. **What is the primary purpose of IoT (Internet of Things)?**

   A. To provide internet access to rural areas
   B. To connect and exchange data between devices
   C. To enhance social media interactions
   D. To improve video streaming quality
   Answer: B. To connect and exchange data between devices

2. **Which of the following is not a common IoT communication protocol?**

   A. Bluetooth
   B. Zigbee
   C. SMTP
   D. Wi-Fi
   Answer: C. SMTP

3. **In an IoT setup, what is the role of a gateway?** A. To act as a central point for data storage
   B. To connect different sensors and devices to the internet
   C. To provide user interfaces for device control
   D. To encrypt data for security purposes
   Answer: B. To connect different sensors and devices to the internet

4. **Which IoT platform is commonly used for cloud-based device management?** A. Windows Server
   B. AWS IoT
   C. Apache Hadoop
   D. Oracle Database
   Answer: B. AWS IoT

5. **Match the IoT term with its correct description:**

IoT devices	**Description**
1. Sensor	A. Executes actions based on received signals
2. Actuator	B. Collects data from the environment

3. Gateway	C. Connects devices to the internet
4. Cloud Platform	D. Provides storage and processing capabilities

Answer:

B: Sensor - Collects data from the environment
2 - A: Actuator - Executes actions based on received signals
3 - C: Gateway - Connects devices to the internet
4 - D: Cloud Platform - Provides storage and processing capabilities

Answer by True if the statement is correct and False if the statement is wrong.

6. IoT devices only operate using Wi-Fi connectivity.
   Answer: False
7. Firmware updates are essential for maintaining the security and functionality of IoT devices. **Answer: True**
8. Cloud platforms are used in IoT systems primarily for data storage and processing.
   Answer: True

Practical assessment

A BUGARAMA Rice Valley Company wants to implement an IoT system for smart irrigation in an agricultural farm to enhance control, conserve water, and improve overall connectivity. Your tasks as technician covers the performance of workplace setup, identification of IoT deployment levels, assembling of IoT equipment, configuration of IoT devices, implementation of IoT system security, and documentation of an IoT installation report. **The duration of the work is 3 hours.** **Resources**	
Tools	Software platforms, simulation software, tape measure, multi-meter, screwdrivers, drilling machine, hammer, ladder, pliers,
Equipment	IoT-enabled Soil Moisture Sensors, Automated Watering System, Centralized IoT Hub, Weather Station Integration and Mobile Application for Monitoring and Control
Material/ Consumable	Internet bundles, power extension, electricity, sensors

SN	Criteria	Indicators	Yes	No
1	Workplace Setup is properly performed	1.1 The workplace environment is well Studied		
		1.2 Requirements Tools, Equipment and Materials are Selected		
		1.3 The workplace is Mapped		
		1.4 Network connectivity is setup		
2	IoT deployment levels are properly identified	2.1 IoT deployment levels are selected		
3	IoT equipment are correctly assembled	3.1 Manufacturer's guide is interpreted		
		3.2 IoT equipment are Fixed		
4	IoT devices are truly configured	4.1 IoT platform is selected		
		4.2 IoT applications are installed		
		4.3 Functional testing are performed		
5	IoT system Security is properly implemented	5.1 The security measures are setup		
		5.2 IoT device firmware Update is performed		
6	IoT installation report is properly generated	6.1 Data are recorded		
		6.2 Report is generated		



Further information to the trainer

"Practical Internet of Things Security" by Brian Russell and Drew Van Duren

"Practical Internet of Things Security" by Brian Russell and Drew Van Duren

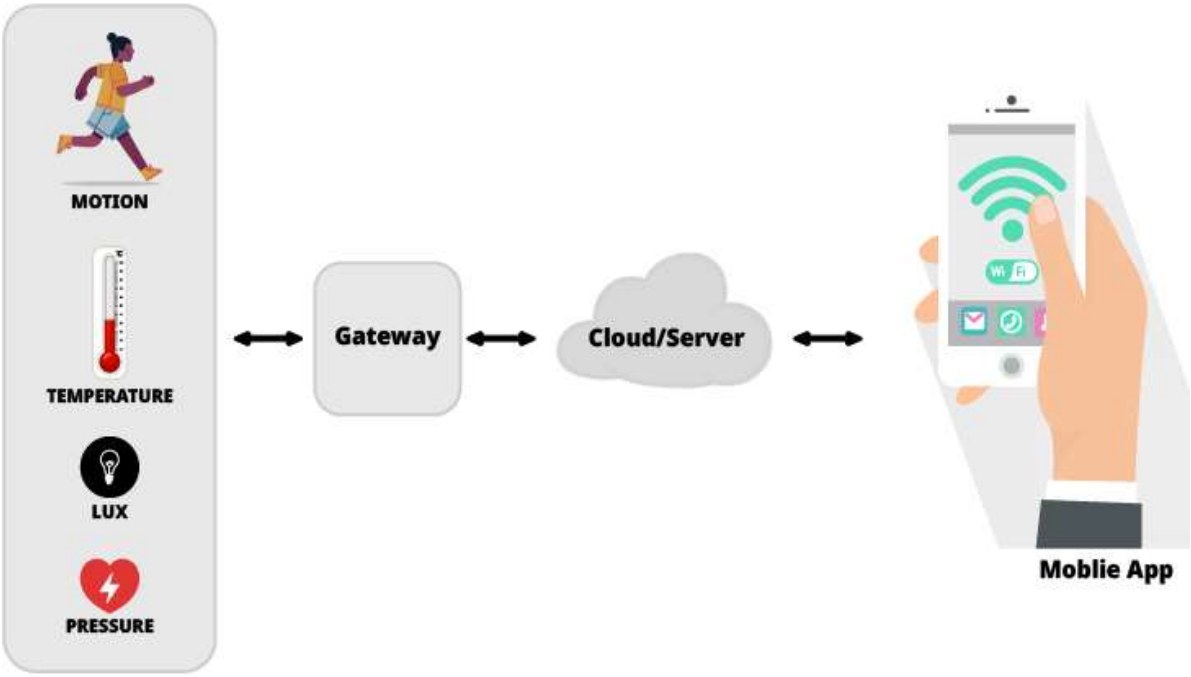
Architecting the Internet of Things,Dieter Uckelmann, Mark Harrison, Michahelles and Florian (Eds), Springer,2011.

Bovell, M.C., Devlin, T.P., Goodner, A.S., Northway, T.N., Usery, P.H.: Wireless deviceconfiguration management. US Patent US8180860B2, 15 May 2012.

Internet-of-Things Device Set Configuration for Connection to Wireless Local Area Network

IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things" by David Hanes, Gonzalo Salgueiro, Patrick Grossetete, and Jerome Henry

Recipes to Begin, Expand, and Enhance Your Projects, 2nd Edition,Michael Margolis,Arduino Cookbook and O‟Reilly Media,2011.


MOTION
TEMPERATURE
LUX
PRESSURE
Gateway
Cloud/Server
Moblie App

Indicative contents
3.1 Identification of IoT systems Types
3.2 Implementation of IoT system Backup and restoration
3.3 Identification of IoT features
3.4 Management of IoT device service
3.5 Monitoring of IoT system

Key Competencies for Learning Outcome 3: Operate IoT System

Knowledge	Skills	Attitudes
• Identification Of Of Iot System Installation Types • Identification Of Iot System Features • Description Of Data Backup • Description of Results Interpretation	• Differentiating Types Of Iot System Installation • Selecting Iot System Features • Creating Data Backup • Monitoring Of Iot System Interpretation	• Having Collaboration • Having a Communication • Having a Problem-Solving • Having a Perseverance Initiative

Duration: 20 hrs

Learning outcome 3 objectives:

By the end of the learning outcome, the trainees will be able to:

1. Identify properly IoT systems Types according to the installation manual
2. Identify properly IoT features according to the IoT system requirements
3. Monitor properly IoT system according to the IoT system requirements
4. Perform correctly Data backup as per the backup guideline
5. Interpret properly IoT system status as per the installation manual
6. Report properly IoT system incidents as per the monitoring results

Resources

Equipment	Tools	Materials
● Modem, ● Access Point, ● Computer ● microcontrollers, ● Arduino Starter Kit	● Software platforms, ● simulation software, ● tape measure, ● multi-meter ● screwdrivers ● drilling machine ● hammer, ● ladder ● pliers	● Internet connectivity ● Pipes ● cable trunks ● electrical tapes ● screws ● Labelling tags

Advance Preparation:

Before delivering this learning outcome, you are recommended to:

- Avail IoT devices, tools, equipment and materials
- Ensure that the installation environment is ready
- Avail resources for participants to further explore Operate IoT system

Indicative content 3.1: Identification of IoT Systems Types

Duration: 4 hrs

Theoretical Activity 3.1.1: Description of Consumer IoT (IoCT) Devices

Notes to the trainer:

- Trainer may use small group for describing Consumer IoT (IoCT) Devices**.** The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. How do they enhance our daily routines?
ii. What is Consumer IoT devices?
iii. Provide examples of CIoT devices.

Step 2: Asks any learner to write their findings on reserved place.
Step 3: Asks learners to discuss the provided answer and choose correct answers.
Step 4: Provides expert view and clarifies ideas by using didactic materials.
Step 5: Address any questions or concerns.
Step 6: Ask trainees to read the key reading 3.1.1 in the trainee manual.

Points to Remember

- **Consumer IoT (CIoT)** refers to the Internet of Things devices used by individuals in their everyday lives. These devices are designed to enhance convenience, entertainment, and health management for consumers. Typically, CIoT solutions leverage Wi-Fi, Bluetooth, and ZigBee to facilitate connectivity. These technologies offer short-range communication suitable for deployments in smaller venues, such as homes and offices.



Theoretical Activity 3.1.2: Description Commercial IoT



Notes to the trainer:

- Trainer may use small group for Describing Commercial IoT (CoIoT) Devices**.** The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions.

i. What is Commercial IoT (CoIoT), and why is it important for different industries?

ii. Can you name one industry where Commercial IoT is commonly used, and provide an example of how it benefits that industry?

Step 2: Asks learners to discuss the provided answer and choose correct answers.

Step 3: Ask students to present their findings to the whole class.

Step 4: Provides expert view and clarifies ideas by using didactic materials.

Step 5: Invite trainees to ask questions and seek clarifications on any aspects of CoIoT use cases discussed during the session.

Step 6: Address any questions or concerns.

Step 7: Ask trainees to read the key reading 3.1.2 in the trainee manual.



Points to Remember

- **Commercial IoT** encompasses IoT devices and systems used in business and commercial settings to improve efficiency, customer experience, and operational management.



Theoretical Activity 3.1.3: Description of Industrial IoT (IoT)



Notes to the trainer:

- Trainer may use small group for Describing Industrial IoT (IIoT) Applications**.** The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. What is IoT, and why is it important in industries?

ii. Can you name one type of industry where IoT is commonly used, and give an example of how it helps that industry?

Step 2: Asks any learner to write their findings on reserved place.

Step 3: Asks learners to discuss the provided answer and choose correct answers.

Step 4: Ask students to present their findings to the whole class.

Step 5: Provides expert view and clarifies ideas by using didactic materials.

Step 6: Address any questions or concerns.

Step 7: Ask trainees to read the key reading 3.1.3 in the trainee manual.



Points to Remember

- **Industrial IoT (IIoT)** involves the use of IoT technologies in industrial sectors such as manufacturing, energy, and logistics. It aims to optimize industrial processes, enhance productivity, and improve safety.



Theoretical Activity 3.1.4: Description of Infrastructure IoT (IIoT)



Notes to the trainer:

- Trainer may use small group for Describing Infrastructure IoT (IIoT). The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

 **Key steps:**

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask learners to perform the task described below
   i. What is Infrastructure IoT (IIoT),
   ii. Why is it important for utilities like water, energy, and transportation systems?

Step 2: Asks any learner to write their findings on reserved place.

Step 3: Asks learners to discuss the provided answer and choose correct answers.

Step 4: Ask students to present their findings to the whole class.

Step 5: Provides expert view and clarifies ideas by using didactic materials.

Step 6: Address any questions or concerns.

Step 7: Ask trainees to read the key reading 3.1.4 in the trainee manual.

 **Points to Remember**

- **Infrastructure IoT** focuses on the use of IoT in infrastructure systems to enhance the efficiency, safety, and sustainability of public services and facilities. Infrastructure IoT is concerned with the development of smart infrastructures that incorporate IoT technologies to boost efficiency, cost savings, maintenance, etc.
- This includes the ability to monitor and control operations of urban and rural infrastructures, such as bridges, railway tracks, and on- and offshore windfarms.

 **Theoretical Activity 3.1.5: Description of Internet of Military Things**

 **Notes to the trainer:**

- Trainer may use small group for Describing Internet of Military Things**.** The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

 **Key steps:**

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:
1. What is the Internet of Military Things (IoMT)?
2. Why is it important for military and defense applications?
3. Can you think of any military or defense situations where technology plays a crucial role? How do you think IoMT might fit into these situations?

Step 2: Asks any learner to write their findings on reserved place.
Step 3: Asks learners to discuss the provided answer and choose correct answers.
Step 4: Provides expert view and clarifies ideas by using didactic materials.
Step 5: Address any questions or concerns.
Step 6: Ask trainees to read the key reading 3.1.5 in the trainee manual.

 **Points to Remember**

- **Internet of Military Things (IoMT)** refers to the application of IoT technologies in military and defense contexts. It aims to enhance situational awareness, operational efficiency, and decision-making capabilities.

 **Application of learning 3.1.**

A mid-sized city is experiencing rapid population growth, leading to increased traffic congestion and longer commute times. The city government has decided to implement a smart traffic management system using IoT to address these challenges. Illustrates how different IoT system types can be integrated to manage a complex urban environment, improving traffic flow, enhancing public safety, and creating a more efficient and responsive city infrastructure.

Checklist:

SN	Criteria	Indicators	Yes	No
1	IoT systems Types are properly identified	1.1 Consumer IoT is described		
		1.2 Commercial IoT is described		
		1.3 Industrial IoT is described		
		1.4 Infrastructure IoT (IIoT) is described		
		1.5 Internet of Military Things (IoMT) is described		

Indicative content 3.2: Implementation of IoT system Backup and Restoration

Duration: 4 hrs

Theoretical Activity 3.2.1: Identification of backup types

Notes to the trainer:

- Trainer may use small group for identifying IoT System Backup. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. What do you understand about IoT system backup and restoration?
ii. Why is Full backup of IoT systems?
iii. Differentiate the differential backup from incremental backup

Step 2: Asks any learner to write their findings on reserved place.
Step 3: Asks learners to discuss the provided answer and choose correct answers.
Step 4: Provides expert view and clarifies ideas by using didactic materials.
Step 5: Address any questions or concerns.
Step 6: Ask trainees to read the key reading 3.2.1 in the trainee manual

Points to Remember

- **Full Back up:** A full backup involves copying all selected data at a specific point in time. It creates a complete copy of the entire dataset, regardless of whether the data has changed since the last backup.

  Characteristics:

  ✓ Captures all data, providing a comprehensive backup.

  ✓ Independent of other backups; each full backup is standalone.

  ✓ Simplifies the restoration process but may consume more storage space.

Use Cases:

- ✓ Initial backup when setting up a new system.
- ✓ Periodic or scheduled backups for critical data.

- **Differential Backup:** A differential backup copies only the data that has changed since the last full backup. It captures the incremental changes from the last full backup, creating a cumulative set of changes since that point.

  Characteristics:

  - ✓ Requires the presence of the last full backup to be effective.
  - ✓ Faster than a full backup, as it only backs up changes since the last full backup.
  - ✓ Restoration involves applying the last full backup and the latest differential backup.

  Use Cases:

  - ✓ Daily or regular backups to capture changes since the last full backup.
  - ✓ Situations where a balance between storage space and backup speed is crucial.

- **Incremental Backup:** An incremental backup captures only the data that has changed since the last backup, whether it's a full or incremental backup. It creates a chain of incremental backups, each building on the previous one.

  Characteristics:

  - ✓ Requires the presence of the last backup (full or incremental) to be effective.
  - ✓ Generally faster and requires less storage space compared to a full or differential backup.
  - ✓ Restoration involves applying the last full backup and all subsequent incremental backups in the correct order.

  Use Cases:

  - ✓ Regular, frequent backups for capturing changes since the last backup.
  - ✓ Situations where storage space is a significant consideration.



Practical Activity 3.2.2: Creation of Backup



Notes to the trainer

- The trainer may use individual based Facilitation of this activity; you are supposed to Create Backup; it is recommended to: Avail examples and exercises on Create Backup and Restoration.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask learners to perform the task described below:
Envision a smart home system that integrates multiple IoT devices, including smart thermostats, security cameras, and home automation controllers. This system gathers information on temperature, security incidents, and user preferences. Considering the importance of this data and the necessity for system reliability, your responsibility is to develop both automated and manual backup procedures for the smart home IoT system.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to calculate the given tasks. While demonstrating, explain the with creating both automatic and manual backup procedures.

Step 4: Give other tasks related to the activity and Asks learners to calculate, monitor the procedures.

Step 5: Verify whether calculations has well done according to the finding answers.

Step 6: Ask trainees to read key reading 3.2.1.

Step 7: Ask trainees to perform the task provided in application of learning 3.2.



Points to Remember

Step 1: Identify Critical Data

Step 2: Choose Backup Types

Step 3: Select Backup Methods

Step 4: Determine Backup Frequency

Step 5: Implement Backup Solutions

Step 6: Verify and Test Backup



Practical Activity 3.2.3: Restoration of IoT system



Notes to the trainer

- The trainer may use individual based Facilitation of this activity; you are supposed to restore IoT system; it is recommended to: Avail examples and exercises on Restoration.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask learners to perform the task described below:
A smart factory uses IoT devices for monitoring and managing production processes. These devices include sensors on machinery, automated conveyor belts, and environmental monitors. The factory relies on real-time data for optimal operation and efficiency. A power surge occurs, causing a network failure and disrupting the connectivity of IoT devices across the factory floor. Several devices lose their configurations, and the central IoT platform stops receiving data. You are assigned to restore the system.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to calculate the given tasks. While demonstrating, explain restoration procedures.

Step 4: Give other tasks related to the activity and Asks learners to calculate, monitor the procedures.

Step 5: Verify whether restorations have well done according to the finding answers.

Step 6: Ask trainees to read key reading 3.2.3.

Step 7: Ask trainees to perform the task provided in application of learning 3.2.



Points to Remember

- **Steps for Restoration of IoT system**

- ✓ Assess the Situation
- ✓ Restore Network Connectivity
- ✓ Device Restoration
- ✓ Data Restoration
- ✓ Application and Software Restoration
- ✓ Security Verification
- ✓ Testing and Validation
- ✓ Documentation and Reporting



Application of learning 3.2.

A smart home system includes various IoT devices such as smart lights, thermostats, security cameras, door locks, and voice assistants. These devices are interconnected and managed via a central hub or cloud platform, providing convenience and automation for the homeowner. A severe storm causes a prolonged power outage, leading to the malfunction of several IoT devices. Some devices lose their settings, and the central hub is unable to connect to the cloud platform. The homeowner needs backup and restore the system to its fully functional state.

Checklist /Solution for application

SN	Criteria	Indicators	Yes	No
1	Backup is planned	Critical Data are identified		
		Backup Types are selected		
		Backup Methods are Selected		
		Backup Frequency is Determined		
		Backup Solutions are Implemented		
		Verify and Test Backups		
2	Restoration is planned	Restoration situation is assessed		
		Network Connectivity is restored		
		Device are restored		
		Data are restored		

		Application and Software are restored		
		Security is verified		
		Testing and Validation are performed		
		Documentation and Reporting are written		

Indicative content 3.3: Identification of IoT features

Duration: 4 hrs

Theoretical Activity 3.3.1: Identification of IoT features

Notes to the trainer:

- Trainer may use small group for identifying IoT features. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:
   i. Enumerate IoT features
   ii. What is the importance for each in IoT system installation?

Step 2: Asks any learner to write their findings on reserved place

Step 3: Asks learners to discuss the provided answer and choose correct answers.

Step 4: Provides expert view and clarifies ideas by using didactic materials.

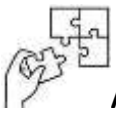
Step 5: Address any questions or concerns.

Step 6: Ask trainees to read the key reading 3.3.1 in the trainee manual.

Points to Remember

- To identify and understand the features of IoT, let's expand on each of the items listed:
  - ✓ **Connectivity**: The ability of devices to connect and communicate with each other and the internet. Enables data exchange and remote control of devices, forming the backbone of IoT systems. They use Wired and wireless communication protocols such as Wi-Fi, Bluetooth, Zigbee, cellular networks (3G, 4G, 5G), LoRa, and more
  - ✓ **Sensing**: The use of sensors to gather data from the physical environment. Provides the necessary data for analysis, monitoring, and automation.

- ✓ **Active Engagements**: The ability for continuous and real-time interaction between devices and users. Ensures timely responses and actions based on real-time data, enhancing user experience and system performance.
- ✓ **Scale**: The capability to handle a large number of devices and vast amounts of data. Allows the IoT system to grow and manage increasing numbers of connected devices and data points. They use Cloud computing, edge computing, and robust data management systems
- ✓ **Dynamic Nature**: The adaptability of IoT systems to changing conditions and environments. Ensures that IoT systems remain functional and relevant even as circumstances evolve.
- ✓ **Intelligence**: The use of advanced algorithms and machine learning to process data and make decisions. Enables predictive analytics, automated decision-making, and smarter system responses. They use Sensors, machine learning algorithms, and context-aware computing
- ✓ **Energy**: Efficient energy management and low power consumption of IoT devices. Extends the lifespan of battery-operated devices and reduces the overall energy footprint of the system. Helps in identifying inefficiencies, optimizing energy usage, and planning for future energy needs.
- ✓ **Safety**: Measures to ensure the safe operation of IoT devices and protect users from harm. Prevents accidents and ensures the reliability and trustworthiness of IoT systems. They use Encryption, authentication protocols, secure hardware, and software updates.
- ✓ **Integration**: The ability to integrate IoT devices with existing systems and technologies. Facilitates seamless operation and enhances the functionality of the entire system by leveraging existing infrastructure. They use APIs, middleware, and integration platforms **c**omponents.



Application of learning 3.3.

Consider a smart building management system that has components includes sensors (temperature, humidity, motion), actuators (lights, HVAC systems), a central controller, cloud storage, and a user interface identify IoT features

SN	Criteria	Indicators	Yes	No
1	IoT features are properly identified	Connectivity is identified		
		Sensing is identified		

		Active Engagements is identified		
		Scale is identified		
		Dynamic Nature is identified		
		Intelligence is identified		
		Energy is identified		
		Safety is identified		
		Integration is identified		

Indicative content 3.4: Management of the IoT devices service

Duration: 4 hrs

Theoretical Activity 3.4.1: Description of Event triggering

Notes to the trainer:

- Trainer may use small group for describing Event for triggering. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. Define Event triggering in the context of computer programming
ii. List example of electronics triggers?

Step 2: Asks any learner to write their findings on reserved place

Step 3: Asks learners to discuss the provided answer and choose correct answers.

Step 4: Provides expert view and clarifies ideas by using didactic materials.

Step 5: Address any questions or concerns.

Step 6: Ask trainees to read the key reading 3.4.1 in the trainee manual.

Points to Remember

- **Event Triggering:** The process of initiating an action or series of actions in response to specific events or conditions detected by IoT devices. Enables real-time responses to changes in the environment, enhancing the functionality and utility of IoT systems.
- **Examples:**
  - ✓ Sensor Data: Triggering an alarm if a smoke detector senses smoke.
  - ✓ Time-Based Events: Turning on lights at a specific time or when it gets dark.
  - ✓ User Actions: Sending a notification when a user enters a geo-fenced area.

- **Benefits of Effective IoT Device Service Management**
    - ✓ Increased Efficiency: Automating responses to events and conditions reduces the need for manual intervention, streamlining operations.
    - ✓ Enhanced User Experience: Tailored actions and configurations improve the usability and convenience of IoT systems.
    - ✓ Cost Savings: Efficient energy management and reduced operational overheads lower costs.
    - ✓ Improved Security: Automated actions enhance security by ensuring timely responses to potential threats.
    - ✓ Scalability: Flexible management systems support the growth and scaling of IoT networks.



Theoretical Activity 3.4.2: Conditional actions



Notes to the trainer:

- Trainer may use small group for Conditional actions. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. Define the conditional actions in context of IoT?
ii. What are the Key Aspects Conditional actions?

Step 2: Asks any learner to write their findings on reserved place
Step 3: Asks learners to discuss the provided answer and choose correct answers.
Step 4: Provides expert view and clarifies ideas by using didactic materials.
Step 5: Address any questions or concerns.
Step 6: Ask trainees to read the key reading 3.4.2 in the trainee manual.



Points to Remember

- **Conditional Actions**: Performing specific actions based on predefined conditions or criteria. Allows for tailored responses and automation, making IoT systems more

intelligent and adaptive to user needs. Utilize rule engines like Node-RED or IFTTT (If This Then That) to define and manage conditional actions.

- **Examples:**
  - ✓ Temperature Control: Adjusting the thermostat if the temperature falls below or exceeds a set threshold.
  - ✓ Security: Locking doors automatically when no movement is detected in a building for a certain period.
  - ✓ Energy Management: Switching off devices when they are not in use to save energy.



Theoretical Activity 3.4.3: Description of Conditional configurations



Notes to the trainer:

- Trainer may use small group for describing Conditional configurations. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:
  i. What mean the conditional configurations in IoT
  ii. What are the examples of Conditional Configurations

Step 2: Asks any learner to write their findings on reserved place
Step 3: Asks learners to discuss the provided answer and choose correct answers.
Step 4: Provides expert view and clarifies ideas by using didactic materials.
Step 5: Address any questions or concerns.
Step 6: Ask trainees to read the key reading 3.4.3 in the trainee manual.



Points to Remember

- **Conditional Configurations**: Changing the configuration or settings of IoT devices based on specific conditions or criteria. Enhances the flexibility and adaptability of IoT systems, ensuring devices operate optimally under varying conditions.

- **Examples**:
    - ✓ Network Settings: Switching to a backup network if the primary connection fails.
    - ✓ Device Behavior: Adjusting the sensitivity of a motion sensor during night hours to reduce false alarms.
    - ✓ Operational Modes: Switching devices to power-saving mode during off-peak hours.



Application of learning 3.4.

For managing a Smart Irrigation System for a Farm. You are requested to automate the irrigation process based on real-time environmental data and predefined conditions to optimize water usage and crop health.

Checklist:

SN	Criteria	Indicators	Yes	No
1	IoT device service is properly managed	Event is triggered		
		Actions are conditioned		
		Conditional is configured		

Indicative content 3.5: Monitoring of IoT system

Duration: 4 hrs

Practical Activity 3.5.1: Monitoring of IoT system software

Notes to the trainer

- The trainer may use individual based facilitation of this activity ; you are supposed to Monitoring of IoT system; it is recommended to: Avail examples and exercises on Monitoring of IoT system**.**

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:
A large commercial building is equipped with a Smart Building Management System (SBMS) to enhance energy efficiency, security, and occupant comfort. The system integrates various IoT devices, including temperature and humidity sensors, smart lighting, HVAC (heating, ventilation, and air conditioning) systems, security cameras, access control systems, and energy meters. Your goal as IoT technician is to monitor the IoT system software to ensure optimal performance, security, reliability, and compliance with building regulations.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to monitor software of the given tasks. While demonstrating, explain Step-by-Step monitoring software procedures

Step 4: Give other tasks related to the activity and Asks learners to select, monitor the procedures.

Step 5: Verify whether monitoring software has well done according to the finding answers.

Step 6: Ask trainees to read key reading **3.5.1**

Step 7: Ask trainees to perform the task provided in application of learning **3.5**

Points to Remember

- IoT System Software Monitoring Checklist:

- ✓ Initial Setup: Verify installations, configurations, and connections.
- ✓ Performance: Monitor CPU, memory, network, set thresholds, review logs.
- ✓ Security: Encrypt data, monitor firewalls, scan vulnerabilities, track access.
- ✓ Reliability: Track uptime, monitor errors, measure MTBF/MTTR, test backups.
- ✓ Health: Monitor battery, signal strength, device status, schedule maintenance.
- ✓ Data Integrity: Ensure consistency, integrity checks, monitor for loss, sync data.
- ✓ Scalability: Monitor resources, handle growth, track limits, plan expansions.
- ✓ Compliance: Audit regularly, meet standards, log inspections, protect data.
- ✓ Incident Response: Set alerts, define protocols, train staff, review plans.
- ✓ Reporting: Generate reports, analyze data, optimize performance, share insights.



Practical Activity 3.5.2: Monitoring of IoT system parameters



Notes to the trainer

- The trainer may use individual based facilitation of this activity ; you are supposed to Monitoring of IoT system parameters; it is recommended to: Avail examples and exercises on Monitoring of IoT system**.**



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:
A large commercial building uses a Smart Building Management System (SBMS) that integrates various IoT devices to enhance energy efficiency, security, and occupant comfort. The system includes temperature and humidity sensors, smart lighting, HVAC systems, security cameras, access control systems, and energy meters. Your task is to monitor the IoT system parameters to ensure energy efficiency, security, and occupant comfort with building regulations.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to monitor software of the given tasks. While demonstrating, explain Step-by-Step monitoring software procedures

Step 4: Give other tasks related to the activity and Asks learners to select, monitor the procedures.

Step 5: Verify whether monitoring system parameters has well done according to the finding answers.

Step 6: Ask trainees to read key reading **3.5.2**

Step 7: Ask trainees to perform the task provided in application of learning **3.5**



Points to Remember

- **Monitoring of IoT system parameters**
  - ✓ Initial Setup: Verify installations, configurations, and connections.
  - ✓ Performance: Monitor CPU, memory, network, set thresholds, review logs.
  - ✓ Security: Encrypt data, monitor firewalls, scan vulnerabilities, track access.
  - ✓ Reliability: Track uptime, monitor errors, measure MTBF/MTTR, test backups.
  - ✓ Health: Monitor battery, signal strength, device status, schedule maintenance.
  - ✓ Data Integrity: Ensure consistency, integrity checks, monitor for loss, sync data.
  - ✓ Scalability: Monitor resources, handle growth, track limits, plan expansions.
  - ✓ Compliance: Audit regularly, meet standards, log inspections, protect data.
  - ✓ Incident Response: Set alerts, define protocols, train staff, review plans.
  - ✓ Reporting: Generate reports, analyze data, optimize performance, share insights.



Theoretical Activity 3.5.3: Interpreting IoT System Status



Notes to the trainer:

- Trainer may use small group for interpreting IoT System Status. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:
  i. Define IoT system status interpretation
  ii. What are some key aspects to consider for IoT system status interpretation?

Step 2: Asks any learner to write their findings on reserved place

Step 3: Asks learners to discuss the provided answer and choose correct answers.

Step 4: Provides expert view and clarifies ideas by using didactic materials.

Step 5: Address any questions or concerns.

Step 6: Ask trainees to read the key reading 3.5.3 in the trainee manual.



Points to Remember

- Interpreting the status of an IoT system involves understanding the complex interplay of its various components and their performance metrics. This is crucial for ensuring system reliability, optimizing performance, and identifying potential issues.
- **Components of IoT System Status**
  - ✓ Device Status
  - ✓ Network Status
  - ✓ Data Status
  - ✓ Application Status
- **Interpretation Techniques**
  - ✓ Threshold-Based Monitoring: Defining predefined limits for system parameters and triggering alerts when exceeded.
  - ✓ Statistical Analysis: Identifying trends, correlations, and anomalies in data patterns.
  - ✓ Machine Learning: Building models to predict system behavior and detect anomalies.
  - ✓ Visualization: Creating visual representations of data to identify patterns and trends.
- **Common IoT System Status Indicators**
  - ✓ Uptime: Percentage of time the system is operational.
  - ✓ Response Time: Time taken for the system to respond to a request.
  - ✓ Error Rate: Frequency of system failures or errors.
  - ✓ Resource Utilization: How efficiently system resources (CPU, memory, storage) are used.
  - ✓ Data Quality: Accuracy and completeness of data.
  - ✓ Security Events: Unauthorized access attempts or breaches.
- **Challenges in Interpreting IoT System Status**
  - ✓ Data Volume: IoT systems generate massive amounts of data, making analysis challenging.
  - ✓ Data Variety: Data comes from different sources with varying formats and structures.
  - ✓ Data Velocity: Data is generated rapidly, requiring real-time processing.
  - ✓ Complex Relationships: System components are interconnected, making it difficult to isolate issues.



Practical Activity 3.5.4: Monitoring results interpretation



Notes to the trainer

- The trainer may use individual based facilitation of this activity; you are supposed Monitoring results interpretation; it is recommended to: Avail examples and exercises on Monitoring of IoT system**.**



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:
You are an IoT technician responsible for monitoring the Smart Building Management System (SBMS) in a large commercial building. The SBMS integrates various IoT devices including temperature and humidity sensors, smart lighting, HVAC systems, security cameras, access control systems, and energy meters. Your goal is to ensure energy efficiency, security, and occupant comfort while complying with building regulations.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to monitor software of the given tasks. While demonstrating, explain Step-by-Step monitoring software procedures

Step 4: Give other tasks related to the activity and Asks learners to select, monitor the procedures.

Step 5: Verify whether monitoring software has well done according to the finding answers.

Step 6: Ask trainees to read key reading **3.5.4**

Step 7: Ask trainees to perform the task provided in application of learning **3.5**



Points to Remember

- **Interpreting Monitoring Results:** Interpreting monitoring results is a critical step in understanding system performance, identifying issues, and making informed decisions. It involves transforming raw data into actionable insights.

Steps of Interpreting Monitoring Results

- **Data Preparation:**
  - ✓ **Cleaning:** Remove errors, inconsistencies, and outliers.

- ✓ **Normalization:** Ensure data is consistent across different sources or time periods.
- ✓ **Aggregation:** Combine data into meaningful summaries (e.g., averages, totals).

Visualization:

- ✓ Graphs and Charts: Create visual representations of data (e.g., line charts, bar charts, histograms).
- ✓ Dashboards: Combine multiple visualizations for a comprehensive overview.
- ✓ Anomaly Detection: Identify unusual data points that may indicate problems.

Analysis:

- ✓ Trend Analysis: Identify patterns and changes over time.
- ✓ Correlation Analysis: Determine relationships between different metrics.
- ✓ Root Cause Analysis: Investigate the underlying reasons for issues or anomalies.

Actionable Insights:

- ✓ Identify Problems: Pinpoint areas requiring attention.
- ✓ Optimize Performance: Make data-driven decisions to improve system efficiency.
- ✓ Predict Future Trends: Anticipate potential issues or opportunities.

- **Common Monitoring Metrics and Their Interpretation**
  - ✓ System Uptime: Measures system availability. High uptime indicates reliability.
  - ✓ Response Time: Indicates how quickly the system responds to requests. Longer response times may signal performance issues.
  - ✓ Error Rates: Measures the frequency of errors. High error rates indicate problems.
  - ✓ Resource Utilization: Tracks CPU, memory, and disk usage. High utilization may indicate bottlenecks.
  - ✓ Network Traffic: Monitors data transfer. Unusual patterns may indicate security threats or performance issues



Theoretical Activity 3.5.6: Describing Incident management



Notes to the trainer:

- Trainer may use small group for describing Incident management. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. Define Incident management
ii. Describe the Lifecycle of Incident management

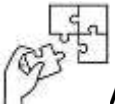
Step 2: Asks any learner to write their findings on reserved place
Step 3: Asks learners to discuss the provided answer and choose correct answers.
Step 4: Provides expert view and clarifies ideas by using didactic materials.
Step 5: Address any questions or concerns.
Step 6: Ask trainees to read the key reading 3.5.3 in the trainee manual.



Points to Remember

- IoT incident management is a structured approach to identifying, responding to, and resolving disruptions or anomalies within an IoT system. It involves detecting issues with IoT devices, networks, or applications, understanding their impact on the overall system, and taking corrective actions to restore normal operations efficiently

- **IoT Incident Management Lifecycle**
  - ✓ Incident Detection:
  - ✓ Incident Triage and Prioritization:
  - ✓ Incident Investigation and Diagnosis:
  - ✓ Incident Resolution:
  - ✓ Incident Closure and Learning



Application of learning 3.5.

You are an IoT technician responsible for monitoring the Smart Building Management System (SBMS) in a large commercial building. The SBMS integrates various IoT devices including temperature and humidity sensors, smart lighting, HVAC systems, security cameras, access control systems, and energy meters. Your task is to monitor the IoT system

Checklist /Solution for application

SN	Criteria	Indicators	Yes	No
1	IoT system is correctly monitored	1.1 software is monitored		
		1.2 Monitoring parameters is monitored		
		1.3 IoT system status are interpreted		
		1.4 Monitoring results are interpreted		
		1.5 Incident is managed		



Learning outcome 3 end assessment

Written assessment

Choose the letter corresponding to the correct answer

1. **Which of the following IoT system types focuses on managing and optimizing public infrastructure?**
    a) Consumer IoT
    b) Commercial IoT
    c) Industrial IoT
    d) Infrastructure IoT

    Answer: d) Infrastructure IoT

2. **What is the main characteristic of an incremental backup?**
    a) Backs up all data every time.
    b) Backs up only data changed since the last full backup.
    c) Backs up only data changed since the last backup, full or incremental.
    d) Backs up only data changed since the last differential backup.

    Answer: c) Backs up only data changed since the last backup, full or incremental.

3. **Which of the following features is associated with IoT devices' ability to collect and process environmental data?**
    a) Connectivity
    b) Sensing
    c) Scale
    d) Safety

    Answer: b) Sensing

Answer by True if the statement is correct and False if the statement is wrong

4. **True or False:** Industrial IoT (IIoT) focuses on enhancing consumer experiences with smart home devices.

    Answer: False

5. **True or False:** Automatic backups are initiated manually by the user or administrator as needed.

Answer: False

6. **True or False:** The "dynamic nature" feature of IoT refers to the system's ability to adapt and reconfigure in real-time based on environmental changes.

   Answer: True

7. **Match the following IoT system types with their primary applications:**

IoT System Type:	**Primary Applications:**
1. Consumer IoT	A. Predictive maintenance and automated production lines
2. Commercial IoT	B. Smart home devices and wearables
3. Industrial IoT	C. Public transportation and smart grids
4. Infrastructure IoT	D. Inventory management and customer analytics
5. Internet of Military Things (IoMT)	E. Surveillance drones and battlefield sensors

Answers: Match the IoT system types with their primary applications:

1 - B (Consumer IoT: Smart home devices and wearables)
2 - D (Commercial IoT: Inventory management and customer analytics)
3 - A (Industrial IoT: Predictive maintenance and automated production lines)
4 - C (Infrastructure IoT: Public transportation and smart grids)
5 - E (Internet of Military Things (IoMT): Surveillance drones and battlefield sensors)

Practical assessment

You have been hired as an IoT systems specialist for a smart manufacturing facility that is implementing a new IoT solution to enhance operational efficiency and safety. The facility includes various IoT devices such as environmental sensors, automated machinery, and a smart energy management system. Your task is to operate and manage the IoT system effectively, ensuring its smooth operation, efficient backup and restoration, and accurate monitoring. **The duration of the work is 3 hours.** **Resources**	
Tools	Software platforms, simulation software, tape measure, multi-meter, screwdrivers, drilling machine, hammer, ladder, pliers,

Equipment	Environmental sensors, automated machinery, and a smart energy management system, Home Automation Hub, Mobile Devices (Smartphones, Tablets) and IoT-enabled Sensors
Material/ Consumable	Internet bundles, power extension, electricity, sensors

Checklist:

SN	Criteria	Indicators	Yes	No
1	IoT system Backup and restoration are correctly implemented	1.1 Backup types identified		
		1.2 backup is created		
		1.3 IoT system is restored		
2	IoT features are properly identified	2.1 IoT features are identified		
3	IoT device service are properly managed	3.1 Event triggering		
		3.2 Conditional actions		
		3.3 Conditional configurations		
4	IoT system are correctly monitored	4.1 Software are Monitored		
		4.2 Parameters are Monitored		
		4.3 IoT system status interpreted		
		4.4 Results interpreted		
		4.5 Incident is managed		

END



Further information to the trainer

"Smart Homes and Their Users" by Springer

"Internet of Things: Concepts, Methodologies, Tools, and Applications" edited by Information Resources Management Association

"Cloud Computing: Concepts, Technology & Architecture" by Thomas Erl, Zaigham Mahmood, and Ricardo Puttini

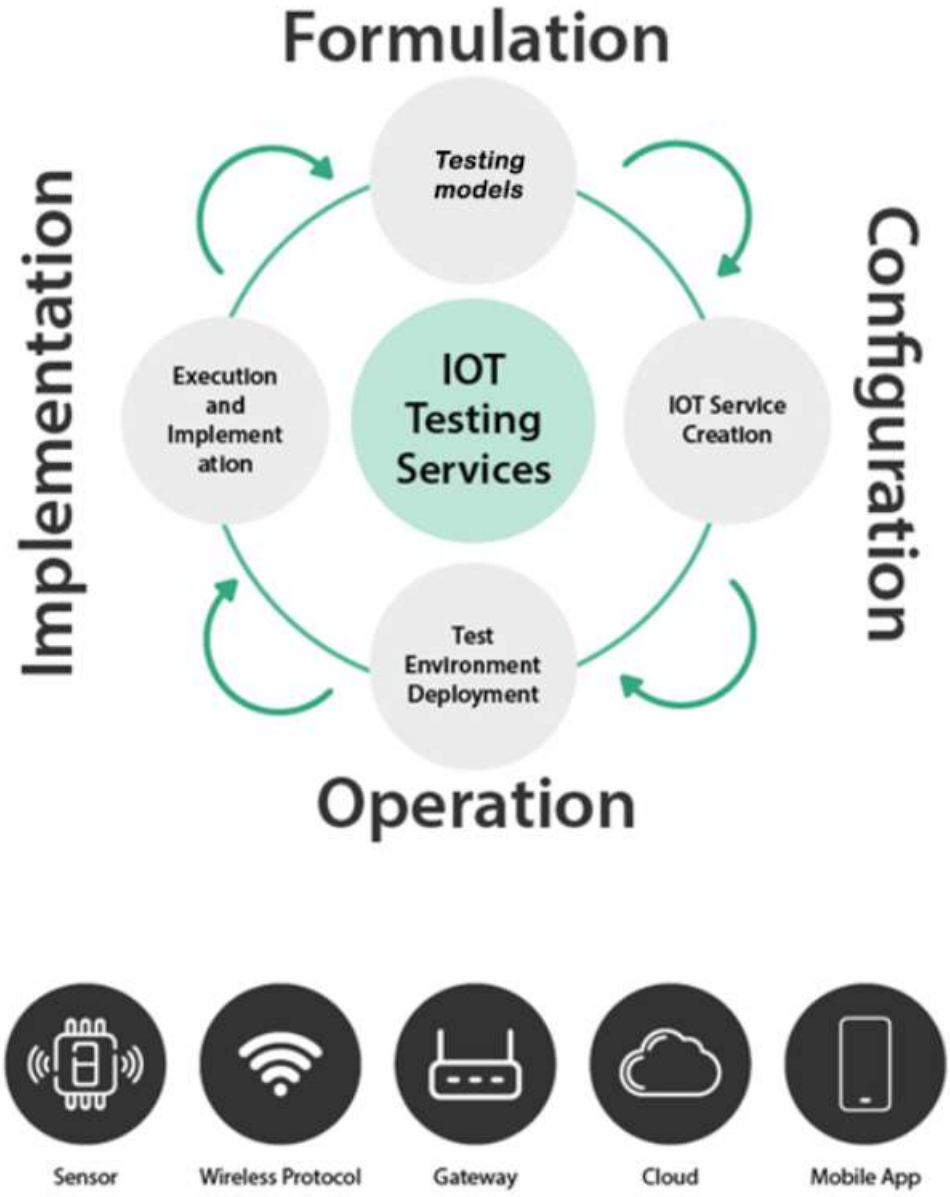
"The Definitive Guide to AWS Infrastructure Automation" by Michael Wittig and Andreas Wittig

"IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things" by David Han

"Designing Connected Products: UX for the Internet of Things" by Claire Rowland, Elizabeth Goodman, and Martin Charlier

"Practical Internet of Things Security" by Brian Russell and Drew Van Duren

"Internet of Things: A Hands-On-Approach" by Arshdeep Bahga and Vijay Madisetti


Formulation
Testing models
Configuration
IOT Service Creation
Test Environment Deployment
Operation
Execution and Implement ation
Implementation
IOT Testing Services
Sensor
Wireless Protocol
Gateway
Cloud
Mobile App

Indicative contents
4.1 Identification of IoT Testing Types
4.2Performance of IoT Tests
4.3 Elaboration of testing report

Key Competencies for Learning Outcome 5: Test IoT system

• Knowledge	• Skills	• Attitudes
• Description of IoT Testing • Performance of IoT Tests • Elaboration Of Testing Report	• Identifying IoT Testing • Selecting of IoT Testing Tools • Generating Of Report	• Having Communication • Having Collaboration • Having Problem-Solving • Having Perseverance Initiative

Duration: 10 hrs

Learning outcome 4 objectives:

By the end of the learning outcome, the trainees will be able to:

1. Describe correctly IoT testing types as used in IoT system installation
2. Perform correctly IoT testing tool as used in IoT system installation
3. Elaborate properly IoT testing report based on IoT system installation

Resources

Equipment	Tools	Materials
● Computer ● Multi-meter ● Digital Storage Oscilloscope	● Software platforms ● JTAG ● Software defined radio (SDR) ● Drilling machine ● hammer, ● ladder ● pliers	● Internet connectivity

Advance Preparation:

Before delivering this learning outcome, you are recommended to:

- Avail IoT devices, tools, equipment and materials
- Ensure that the installation environment is ready
- Avail resources for participants to further explore Test IoT system

Indicative content 4.1: Identification of IoT Testing Types

Duration: 3 hrs

Theoretical Activity 4.1.1: Description of IoT Testing Types

Notes to the trainer:

- Trainer may use small group for describing IoT Testing Types. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. What is Usability in IoT Testing?
ii. Define Reliability in IoT?
iii. Explain Security in IoT testing?
iv. What is the difference between Data integrity and Compatibility
v. Mention applications of Performance in IoT

Step 2: Asks any learner to write their findings on reserved place.
Step 3: Asks learners to discuss the provided answer and choose correct answers.
Step 4: Provides expert view and clarifies ideas by using didactic materials.
Step 5: Address any questions or concerns.
Step 6: Ask trainees to read the key reading 4.1.1 in the trainee manual.

Points to Remember

- **Usability Testing:** Ensuring the IoT device or application is user-friendly and intuitive. Key areas: User interface design, Device ergonomics and User experience across different platforms (mobile, web, etc.).
- **Reliability Testing:** Evaluating the system's ability to perform its intended function without failure over time**.** Key areas: Mean time between failures (MTBF), Device lifespan and System recovery capabilities.
- **Data Integrity Testing:** Ensuring data accuracy, completeness, and consistency throughout the IoT system. Key areas: Data validation and cleansing, Data encryption and protection, Data synchronization and consistency.

- **Security Testing:** Identifying vulnerabilities and weaknesses in the IoT system to protect against threats. Key areas: Network security, Device security, Data privacy and protection, Authentication and authorization.
- **Compatibility Testing:** Verifying the IoT system's ability to interoperate with different hardware, software, and network environments. Key areas: Device compatibility, Network compatibility, Platform compatibility and Software compatibility.
- **Performance Testing:** Evaluating the system's response time, throughput, scalability, and stability under various workloads**.** Key areas: Load testing, Stress testing and Endurance testing.



Application of learning 4.1

Perform the testing the security of a Smart Home System to ensure that the smart home system is secure against unauthorized access and data breaches.

Checklist:

SN	Criteria	Indicators	Yes	No
1	IoT Testing Types are properly identified	1.1Usability testing is identified		
		1.2 Reliability testing is identified		
		1.3 Data integrity testing is identified		
		1.4 Security testing is identified		
		1.5 Compatibility testing is identified		
		1.6 Performance testing is identified		

Indicative content 4.2: Performance of IoT Tests

Duration: 4 hrs

Practical Activity 4.2.1: Perform Functional test

Notes to the trainer

- The trainer may use individual based facilitation of this activity; you are supposed Functional test; it is recommended to: Avail examples and exercises on Functioning test of IoT system**.**

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:

You are conducting a test on a smart home lighting system for a customer who wishes to confirm that their smart light bulbs can be operated via a mobile application. Perform Functional test to validate the essential functions of the smart home lighting system

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)
Step 3: Demonstrate how to Functional test of the given tasks. While demonstrating, explain step-by-step Functional test procedures
Step 4: Give other tasks related to the activity and Asks learners to select, monitor the procedures.
Step 5: Verify whether functional test has well done according to the finding answers.
Step 6: Ask trainees to read key reading 4.2.1

Points to Remember

Step 1: Test Environment Setup:

- ✓ Hardware Setup: Connect the smart bulb and Hue Bridge to power sources.
- ✓ Software Setup: Install and configure the Hue App on a smartphone.

- ✓ Network Configuration: Ensure the Hue Bridge is connected to the home Wi-Fi network.

Step 2: Execution of Functional Tests:

- ✓ **TC-01 Execution:**
  - Step 1: Power on the smart bulb.
  - Step 2: Ensure the Hue Bridge is connected to the Wi-Fi.
  - Step 3: Use the Hue App to connect the bulb to the Bridge.
  - Expected Result: The bulb connects successfully and appears in the app.
  - Actual Result: (Record the outcome after execution)
- ✓ **TC-02 Execution:**
  - Step 1: Open the Hue App.
  - Step 2: Turn the bulb on/off using the app.
  - Step 3: Change the bulb color.
  - Step 4: Adjust the brightness.
  - Expected Result: The bulb responds accurately to all commands.
  - Actual Result: (Record the outcome after execution)
- ✓ **TC-03 Execution:**
  - Step 1: Add multiple bulbs to the Bridge using the app.
  - Step 2: Group the bulbs in the app.
  - Step 3: Control the group (turn on/off, change color, adjust brightness).
  - Expected Result: All bulbs in the group respond simultaneously to controls.
  - Actual Result: (Record the outcome after execution)

Step 3: Data Collection and Analysis:

- ✓ Logs and Reports: Capture logs from the Hue App during the test.
- ✓ Issue Tracking: Note any issues or discrepancies found during testing and document them for further investigation.



Practical Activity 4.2.2: Performing Security test



Notes to the trainer

- The trainer may use individual based facilitation of this activity; you are supposed to perform Security test; it is recommended to: Avail examples and exercises on performing Security test of IoT system**.**



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:

You are conducting a test on a smart home lighting system for a customer who wishes to confirm that their smart light bulbs can be operated via a mobile application. The customer has set up Philips Hue smart bulbs throughout their home, connected to a Hue Bridge, and uses the Hue mobile app for control. The objective is to validate the essential functions of the smart home lighting system, ensuring that the smart bulbs can connect to the Hue Bridge and be controlled through the Hue mobile app.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to Functional test of the given tasks. While demonstrating, explain step-by-step Functional test procedures

Step 4: Give other tasks related to the activity and Asks learners to select, perform Security test the procedures.

Step 5: Verify whether security test has well done according to the finding answers.

Step 6: Ask trainees to read key reading 4.2.2



Points to Remember

- **Step1:** Identification of Security Testing Tools
  - ✓ OWASP ZAP: An open-source web application security scanner.
  - ✓ Nmap: A network scanning tool to discover hosts and services on a computer network.

- ✓ Wireshark: A network protocol analyzer to capture and analyze network traffic.
- ✓ Burp Suite: An integrated platform for performing security testing of web applications.
- ✓ Nessus: A vulnerability scanner to identify potential vulnerabilities in the system.
- ✓ Metasploit: A penetration testing framework for discovering, exploiting, and validating vulnerabilities.

- **Step2: Perform Security Test**
  Test Case: Penetration Testing on IoT Device

  Test Steps:

  - ✓ Nmap: Perform a network scan to identify open ports and services running on the IoT device. Command: nmap -sV <IoT_device_IP>
  - ✓ Wireshark: Capture network traffic to analyze communication between the IoT device and other devices.
  - ✓ Start capturing packets, filter by the IoT device's IP, and analyze the traffic.
  - ✓ OWASP ZAP: Conduct an automated scan for web application vulnerabilities if the IoT device has a web interface.
  - ✓ Set up a proxy, configure the target, and initiate the scan.
  - ✓ Nessus: Perform a vulnerability scan on the IoT device.
  - ✓ Configure a scan policy, add the target, and run the scan.
  - ✓ Metasploit: Attempt to exploit identified vulnerabilities to test their impact. Command: use <exploit_name> and follow the framework instructions.



Practical Activity 4.2.3: Perform Connectivity test



Notes to the trainer

- The trainer may use individual based facilitation of this activity; you are supposed to perform Connectivity test; it is recommended to: Avail examples and exercises on performing Connectivity test of IoT system**.**

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:

You are conducting a test on a smart home lighting system for a customer who wishes to confirm that their smart light bulbs can be operated via a mobile application. The customer has set up Philips Hue smart bulbs throughout their home, connected to a Hue Bridge, and uses the Hue mobile app for control. The objective is to validate the essential functions of the smart home lighting system, ensuring that the smart bulbs can connect to the Hue Bridge and be controlled through the Hue mobile app.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to perform connectivity test of the given tasks. While demonstrating, explain step-by-step Connectivity test procedures

Step 4: Give other tasks related to the activity and Asks learners to select connectivity test procedures.

Step 5: Verify whether connectivity test has well done according to the finding answers.

Step 6: Ask trainees to read key reading 4.2.3

Step 7: Ask trainees to perform the task provided in application of learning 4.2

Points to Remember

Step1: Identification of Connectivity Testing Tools

- ✓ Ping: A basic tool to test connectivity between devices.
- ✓ Traceroute: A tool to trace the path packets take to reach a destination.
- ✓ Wireshark: For capturing and analyzing network traffic.
- ✓ iPerf: A tool to measure network bandwidth.
- ✓ Netcat: A utility to test network connections and port communication.

Step2: Perform Connectivity Test

- ✓ Test Steps:
  - Ping: Test basic connectivity to the IoT device.
    - Command: ping <IoT_device_IP>

- Traceroute: Trace the route packets take to reach the IoT device.
    - Command: traceroute <IoT_device_IP>
- Wireshark: Capture and analyze network traffic to ensure proper communication.
    - Steps: Start capturing packets, filter by the IoT device's IP, and analyze the traffic.
- iPerf: Measure network bandwidth between the IoT device and another host.
    - Command: iperf -c <host_IP> -t 60
- Netcat: Test port communication and ensure services are accessible.
    - Command: nc -zv <IoT_device_IP> <port>

- Expected Result: IoT device should have stable and reliable connectivity with no packet loss or significant delays.
- Actual Result: *(To be recorded during the test)*



Practical Activity 4.2.4: Perform Performance test



Notes to the trainer

- The trainer may use individual based facilitation of this activity; you are supposed to perform Performance test; it is recommended to: Avail examples and exercises on performing Performance test of IoT system**.**



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:

You are conducting a test on a smart home lighting system for a customer who wishes to confirm that their smart light bulbs can be operated via a mobile application. The customer has set up Philips Hue smart bulbs throughout their home, connected to a Hue Bridge, and uses the Hue mobile app for control. The objective is to validate the essential functions of the smart home lighting system,

ensuring that the smart bulbs can connect to the Hue Bridge and be controlled through the Hue mobile app.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)
Step 3: Demonstrate how to Performance test of the given tasks. While demonstrating, explain step-by-step Performance test procedures
Step 4: Give other tasks related to the activity and Asks learners to select, performing test the procedures.
Step 5: Verify whether performing teshas well done according to the finding answers.
Step 6: Ask trainees to read key reading 4.2.4
Step 7: Ask trainees to perform the task provided in application of learning 4.2



Points to Remember

Step1: **Identification of Performance Testing Tools**

- ✓ Apache JMeter: A tool to measure performance and load.
- ✓ Gatling: A tool for load and performance testing.
- ✓ Locust: An open-source load testing tool.
- ✓ Wireshark: To analyze network performance.
- ✓ iPerf: To measure network bandwidth and performance.

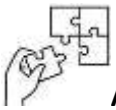
Step2: Perform Performance Test

Test Case: Load Testing on IoT System

- ✓ Objective: To assess the performance of the IoT system under various load conditions.
- ✓ Preconditions: IoT system is set up and connected to the network, and performance testing tools are installed.
- ✓ Test Steps:
  - Apache JMeter: Configure a test plan to simulate multiple users interacting with the IoT system.
    - Steps: Create a thread group, add HTTP requests or MQTT samplers, configure listeners, and run the test.
  - Gatling: Create and execute a simulation to load test the IoT system.
    - Steps: Write a simulation script, configure the load, and execute the test.

- Locust: Simulate multiple users to test the system's load handling capabilities.
    - Steps: Write a task script, configure the number of users, and start the test.
- Wireshark: Capture and analyze network traffic to identify any performance bottlenecks.
    - Steps: Start capturing packets, filter by the IoT device's IP, and analyze the traffic.
- iPerf: Measure the network bandwidth and latency under load conditions.
    - Command: iperf -c <host_IP> -t 60

- Expected Result: The IoT system should handle the load without significant performance degradation or failures.
- Actual Result: *(To be recorded during the test)*



Application of learning 4.2

A smart factory is implementing an IoT-based monitoring system to oversee the performance of IoT Tests of industrial machines. The system involves sensors attached to each machine, which gather data on temperature, vibration, energy consumption, and operational status in real time. Your tasks are to perform the IoT Tests.

SN	Criteria	Indicators	Yes	No
1	IoT Tests are properly performed	Functional test is performed		
		Security test is performed		
		Connectivity test is performed		
		Performance test is performed		

Indicative content 4.3: Elaboration of Testing Report

Duration: 3 hrs

Practical Activity 4.3.1: Generation of report

Notes to the trainer

- The trainer may use individual based facilitation of this activity; you are supposed to generate of report; it is recommended to: Avail examples and exercises on performing Performance test of IoT system**.**

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:
You are testing a smart home lighting system. As an IoT Technician, you are required to generate a report documenting all the tests conducted.
Step 2: Explain the task and provide clear work instruction (Task, Time allocated)
Step 3: Demonstrate how to generate a report of the given tasks. While demonstrating, explain step-by-step generation of report procedures
Step 4: Give other tasks related to the activity and Asks learners to select, generate a report the procedures.
Step 5: Verify whether generating a report has well done according to the finding answers.
Step 6: Ask trainees to read key reading 4.3.1
Step 7: Ask trainees to perform the task provided in application of learning 4.3

Points to Remember

- **Generation of Report**
  The primary objective of generating a testing report is to document the results of various tests conducted on IoT systems. This report provides a comprehensive overview of the system's performance, reliability, and security.
  Components:

- **Title Page:**
  - ✓ Title of the Report
  - ✓ Project Name
  - ✓ Date of Report
  - ✓ Author/Tester Name
- **Table of Contents:** A list of sections and their respective page numbers for easy navigation.
- **Executive Summary:** A brief summary of the testing process, key findings, and overall assessment of the system.
- **Introduction:**
  - ✓ Purpose of the Report
  - ✓ Scope of Testing
  - ✓ Description of the IoT system under test
- **Testing Methodology:**
  - ✓ Types of Tests Conducted (e.g., functional, performance, security, usability)
  - ✓ Testing Environment (hardware and software used)
  - ✓ Test Cases and Test Scenarios
- **Test Execution Details:**
  - ✓ Schedule and timeline of the tests
  - ✓ Test execution logs
  - ✓ Any deviations from the planned testing process
- **Test Results:**
  - ✓ Summary of test outcomes (pass/fail status)
  - ✓ Detailed results for each test case
  - ✓ Screenshots or logs illustrating test results
- **Defects and Issues:**
  - ✓ List of identified defects
  - ✓ Severity and priority of each defect
  - ✓ Status of defects (open, resolved, in progress)

- ✓ Steps to reproduce defects

- **Performance Metrics:**
  - ✓ System response time
  - ✓ Throughput
  - ✓ Resource utilization (CPU, memory, network)
- **Security Findings:**
  - ✓ Vulnerability assessment results
  - ✓ Penetration testing outcomes
  - ✓ Recommendations for addressing security issues
- **Conclusion:**
  - ✓ Overall assessment of the system
  - ✓ Summary of key findings
  - ✓ Recommendations for improvements
- **Appendices:** Additional data, logs, test scripts, and any other relevant documentation



Practical Activity 4.3.2: Interpretation of report



Notes to the trainer

- The trainer may use individual based facilitation of this activity; you are supposed to Interpret of report; it is recommended to: Avail examples and exercises on Interpretation of report test of IoT system**.**



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:

You are testing a smart home lighting system. As an IoT Technician, you are required to interpret the report generated from the tests conducted

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to interpret a report of the given tasks. While demonstrating, explain step-by-step interpretation of report procedures

Step 4: Give other tasks related to the activity and Asks learners to select, monitor the procedures.

Step 5: Verify whether interpreting of report has well done according to the finding answers.

Step 6: Ask trainees to read key reading 4.3.2

Step 7: Ask trainees to perform the task provided in application of learning 4.3



Points to Remember

- **Interpretation of Report:** The interpretation of the testing report involves analyzing the data and findings to derive meaningful insights about the system's performance and areas for improvement
- **Steps:**
  - **Review Executive Summary:** Understand the high-level outcomes and overall assessment provided by the tester.
  - **Analyze Test Results:** Examine the detailed results for each test case and identify patterns or recurring issues that may indicate underlying problems.
  - **Evaluate Defects and Issues:** Assess the severity and impact of identified defects and prioritize defects based on their criticality and impact on the system.
  - **Assess Performance Metrics:** Compare performance metrics against predefined benchmarks or requirements and identify any bottlenecks or performance degradation.
  - **Examine Security Findings:** Analyze vulnerabilities and security weaknesses and determine the potential risks and their implications for the system.
  - **Draw Conclusions:** Synthesize findings from different sections of the report and determine the overall health and readiness of the system for deployment.
  - **Provide Recommendations:** Suggest corrective actions for identified issues and recommend best practices for improving system performance and security.

- **Prepare Action Plan:** Develop a plan for addressing the identified issues and assign responsibilities and set timelines for resolution.



Application of learning 4.2

You are testing a smart home lighting system. As an IoT Technician, you are required to generate and interpret the report from the tests conducted.

SN	Criteria	Indicators	Yes	No
1	Testing report is properly elaborated	report is generated		
		Report is interpreted		



Learning outcome 4 end assessment

Written assessment

Choose the letter corresponding to the correct answer:

1. **Which of the following is NOT a type of IoT test?**

   A) Usability

   B) Compatibility

   C) Functionality

   D) Scalability

2. **Which tool is commonly used for security testing in IoT systems?**

   A) JMeter

   B) Wireshark

   C) LoadRunner

   D) OWASP ZAP

3. **What is the primary purpose of a connectivity test in IoT systems?**

   A) To evaluate the system's responsiveness

   B) To ensure devices can connect and communicate reliably

   C) To verify the accuracy and consistency of data

   D) To check for vulnerabilities and ensure the system is protected against threats

4. **True or False:** Usability testing in IoT systems assesses how user-friendly and intuitive the system is.

5. **True or False:** JMeter is a tool used primarily for security testing in IoT systems.

6. **True or False:** Connectivity tests ensure the system performs consistently under specified conditions.

7. Match the IoT testing type to its primary focus

IoT testing	**Its primary focus**
Usability	A) Ensures the system performs consistently under specified conditions
Reliability	B) Assesses how user-friendly and intuitive the system is
Data Integrity	C) Evaluates the system's responsiveness, throughput, and scalability
Security	D) Verifies the accuracy and consistency of data over its lifecycle
Compatibility	E) Checks for vulnerabilities and ensures the system is protected against threats
Performance	F) Tests the system's ability to function with different devices and environments

Answers

1. **Which of the following is NOT a type of IoT test?**

   D) Scalability

2. **Which tool is commonly used for security testing in IoT systems?**

   D) OWASP ZAP

3. **What is the primary purpose of a connectivity test in IoT systems?**

   B) To ensure devices can connect and communicate reliably

4. **True or False:** Usability testing in IoT systems assesses how user-friendly and intuitive the system is.

   True

5. **True or False:** JMeter is a tool used primarily for security testing in IoT systems.

   False (JMeter is primarily used for performance testing)

6. **True or False:** Connectivity tests ensure the system performs consistently under specified conditions.

   False (Connectivity tests ensure devices can connect and communicate reliably)

7. **Match the IoT testing type to its primary focus:**

   1. Usability - B) Assesses how user-friendly and intuitive the system is

2. Reliability - A) Ensures the system performs consistently under specified conditions
3. Data Integrity - D) Verifies the accuracy and consistency of data over its lifecycle
4. Security - E) Checks for vulnerabilities and ensures the system is protected against threats
5. Compatibility - F) Tests the system's ability to function with different devices and environments
6. Performance - C) Evaluates the system's responsiveness, throughput, and scalability

Practical assessment

As a testing professional, your responsibility is to ensure the reliability and effectiveness of an Industrial IoT system designed for predictive maintenance in a manufacturing facility. The system incorporates sensors on machinery, data analytics tools, and a central monitoring platform. The primary objective is to identify potential issues in the machinery before they lead to failures, optimizing maintenance efforts, and minimizing downtime. Your tasks include the identification of IoT testing types, the performance of IoT tests, and the elaboration of a comprehensive testing report.

The duration of the work is 3 hours.

Resources

Tools	Software platforms, simulation software, tape measure, multi-meter, screwdrivers, drilling machine, hammer, ladder, pliers,
Equipment	Smart Lighting System, Smart Thermostats, Smart Security Cameras, Home Automation Hub, Mobile Devices (Smartphones, Tablets) and IoT-enabled Sensors
Material/ Consumable	Internet bundles, power extension, electricity, sensors

Checklist:

SN	Criteria	Indicators	Yes	No
1	IoT Testing Types are properly identified	1.1 Usability is identified		
		1.2 Reliability is identified		
		1.3 Data integrity is identified		
		1.4 Security is identified		
		1.5 Compatibility is identified		
		1.6 Performance is identified		
2	IoT Tests are properly performed	2.1 Functional test is Performed		
		2.2 Security test is Performed		
		2.3 Connectivity test is Performed		
		2.4 Performance test is Performed		
3	Testing report is properly elaborated	4.1 The performance of individual sensors is recorded and analysed		
		3.2 Report is generated		
		3.3 Report is interpreted		

END



Further information to the trainer

"Foundations of Software Testing: ISTQB Certification" by Rex Black, Erik van Veenendaal, and Dorothy Graham

"Internet of Things: A Hands-On-Approach" by Arshdeep Bahga, Vijay Madisetti

"Internet of Things: Principles and Paradigms" by Rajkumar Buyya, Amir Vahid Dastjerdi

"Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die" by Eric Siegel

"Sensor Technologies: Healthcare, Wellness and Environmental Applications" by Michael J. McGrath

"Software Testing: A Craftsman's Approach" by Paul C. Jorgensen

"Wireless Sensor Networks: Technology, Protocols, and Applications" by Kazem Sohraby, Daniel Minoli, and Taieb Znati



Indicative contents
5.1 Identification of IoT Maintenance types
5.2 Application of IoT Software upgrade
5.3 Performance of IoT system security maintenance
5.4 Documentation of IoT maintenance report

Key Competencies for Learning Outcome 5: Maintain IoT System

Knowledge	Skills	Attitudes
● Identification of IoT Maintenance types ● Application of IoT Software upgrade ● Performance of IoT system security maintenance ● Documentation of IoT maintenance report	● Applying Of IoT Software Upgrade ● Performing IoT Action Repairs ● Performing IoT System Protection Checks ● Generating Report	● Having a Communication And Collaboration ● Having a Problem-Solving ● Having a Perseverance Initiative ● Having a Time Management

Duration: 10 hrs

Learning outcome 5 objectives:

By the end of the learning outcome, the trainees will be able to:

1. Identify correctly IoT testing types as used in IoT system installation
2. Apply correctly IoT Software upgrade as used in IoT system
3. Perform properly IoT system security maintenance based on IoT system installation
4. Document correctly IoT maintenance report based on IoT system installation

Resources

Equipment	Tools	Materials
● Computer ● Multi-meter ● Digital Storage Oscilloscope	● Software platforms ● JTAG ● Software defined radio (SDR) ● Drilling machine ● hammer, ● ladder ● pliers	● Internet connectivity

Advance Preparation:

Before delivering this learning outcome, you are recommended to:

- Avail IoT devices, tools, equipment and materials
- Ensure that the installation environment is ready
- Avail resources for participants to further explore maintain IoT system

Indicative content 5.1: Identification of IoT Testing Types

Duration: 2 hrs

Theoretical Activity 5.1.1: Identification of IoT Maintenance types

Notes to the trainer:

- Trainer may use small group for identifying IoT Maintenance types. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. Define IoT maintenance
ii. Differentiate predictive from preventive maintenance in IoT and provide examples?

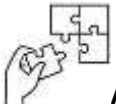
Step 2: Asks any learner to write their findings on reserved place.
Step 3: Asks learners to discuss the provided answer and choose correct answers.
Step 4: Provides expert view and clarifies ideas by using didactic materials.
Step 5: Step 5: Address any questions or concerns.
Step 6: Step 6: Ask trainees to read the key reading 5.1.1 in the trainee manual.

Points to Remember

- **IoT maintenance** refers to the process of ensuring the optimal functionality, security, and efficiency of interconnected devices and systems within the Internet of Things (IoT) ecosystem.
- **Identification of IoT Maintenance types**
  - **Preventive Maintenance:**
    - ✓ **Regular Inspections:** Periodic checks and inspections to identify potential issues before they escalate.
    - ✓ **Scheduled Maintenance:** Planned maintenance activities based on predetermined schedules to ensure device longevity.

- ✓ **Software Updates:** Applying firmware and software updates to enhance features, security, and performance.

- **Predictive Maintenance:**
  - ✓ **Data Analytics:** Utilizing real-time data and analytics to predict when equipment failure or maintenance is likely to occur, allowing for proactive interventions.
  - ✓ **Machine Learning Algorithms:** Employing machine learning models to analyse historical data and predict future maintenance needs based on patterns and trends.
- **Corrective Maintenance:**
  - ✓ Fault Detection: Identifying and diagnosing malfunctions or failures in real-time to promptly initiate corrective actions.
  - ✓ Repairs and Replacements: Conducting repairs, component replacements, or device replacements to restore functionality.



Application of learning 5.1.

An HVAC (heating, ventilation, and air conditioning) system in a smart manufacturing plant is monitored and controlled by means of a temperature sensor that is part of the smart manufacturing plant systems. Describe different types of IoT Maintenance you can apply to ensure the continuous and efficient operation of a smart manufacturing plant using various IoT maintenance strategies.

Checklist for application

SN	Criteria	Indicators	Yes	No
1	IoT Maintenance are properly identified	1.1 Predictive Maintenance is described		
		1.2 Preventive is described		
2	IoT tools are properly used	2.1 Hardware tools are selected and cleaned		
		2.2 Software platforms tools are selected		

Indicative content 5.2: Application of IoT Software upgrade

Duration: 3 hrs

Theoretical Activity 5 .2.1: Application of IoT Software upgrade

Notes to the trainer:

- Trainer may use small group for identifying IoT Maintenance types. The use of images, videos, and illustrations as didactic materials is required.
- Avail tools, materials and equipment related to this activity.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. Define a software upgrade?

ii. Differentiate software upgrade and update.

iii. What are the types of software upgrade?

iv. How is the software upgrade is done? (list and explain options)

Step 2: Asks trainees to write provided answers on flipchart/paper.

Step 3: Asks trainees to present their findings to the whole class.

Step 4: Provides expert view and Address any questions or concerns.

Step 5: Ask trainees to read the key reading 5.2.1 in the trainee manual.

Points to Remember

- **Automatic Software Upgrades**
  - Advantages:
    - ✓ Efficiency: Updates can be deployed across multiple devices simultaneously, saving time and reducing the need for manual intervention.

- ✓ Consistency: Ensures all devices are running the latest software version, reducing discrepancies and potential compatibility issues.
- ✓ Security: Automatic updates can quickly address security vulnerabilities, reducing the risk of attacks or exploits.
- ✓ User Experience: Minimal disruption to users, as updates can be scheduled during off-peak hours or pushed silently in the background.

Considerations:

- ✓ Control: Users or administrators may have limited control over when updates are applied, which could be an issue in critical or sensitive environments.
- ✓ Testing: Updates should be thoroughly tested before deployment to avoid introducing new bugs or issues across all devices.
- ✓ Bandwidth: Large updates can consume significant bandwidth, potentially affecting network performance.

- **Manual Software Upgrades**

Advantages:

- ✓ Control: Administrators can decide when to apply updates, allowing for planned downtime and minimal disruption.
- ✓ Customization: Allows for selective updating of devices, which can be useful if certain devices need to remain on a specific version for compatibility reasons.
- ✓ Testing: Updates can be tested on a small subset of devices before being rolled out more widely, reducing the risk of widespread issues.

Considerations:

- ✓ Time-Consuming: Manually updating devices can be labor-intensive, especially in large-scale IoT deployments.
- ✓ Inconsistency: Higher risk of some devices being missed or not updated promptly, leading to potential security vulnerabilities and compatibility issues.
- ✓ User Dependency: Relies on the diligence and availability of administrators or users to perform updates, which can lead to delays.

- **Implementation Strategies**

Automatic Upgrades:

- ✓ Over-the-Air (OTA) Updates: Commonly used in IoT devices, where updates are pushed to devices wirelessly.
- ✓ Scheduled Updates: Automatic updates can be scheduled during off-peak hours to minimize disruption.
- ✓ Rolling Updates: Gradually updating a subset of devices at a time to monitor for any issues before a full rollout.

Manual Upgrades:

- ✓ Local Updates: Using USB drives or other local media to manually update each device.
- ✓ Centralized Management Systems: Administrators use centralized systems to manage and deploy updates to specific devices as needed.
- ✓ User-Initiated Updates: End users are notified and given instructions on how to manually update their devices.



Application of learning 5.2.

A smart home system integrates various IoT devices, including smart thermostats, security cameras, lighting controls, door locks, and voice assistants. These devices communicate via a central hub and are controlled through a mobile app or voice commands. The system is widely used by homeowners to enhance comfort, security, and energy efficiency. Implement a software upgrade for a smart home system to enhance security, add new features, and improve system performance without disrupting user experience

Checklist for application

SN	Criteria	Indicators	Yes	No
1	IoT software tools are correctly selected	1.1 Software platforms are selected		
		1.2 Software tools are selected		
2	IoT Software upgrade is properly applied	2.1 Manual IoT Software upgrade is applied		
		2.2 Automatic IoT Software upgrade is applied		

Indicative content 5.3: Performance of IoT system security maintenance

Duration: 3 hrs

Practical Activity 5.3.1: Performing IoT action repairs

Notes to the trainer

- The trainer may use individual based facilitation of this activity; you are supposed to perform IoT action repairs; it is recommended to: Avail examples and exercises on performing IoT action repairs.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:
A security camera on your IoT network is found to be transmitting unusual amounts of data at unexpected times, indicating a potential breach or malfunction. As an IoT Technician you are asked to repair issue.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to Perform IoT action repairs of the given tasks. While demonstrating, explain step-by-step Performing IoT action repairs procedures

Step 4: Give other tasks related to the activity and Asks learners to select, Performing IoT action repairs the procedures.

Step 5: Verify whether Performing IoT action repairs has well done according to the finding answers.

Step 6: Ask trainees to read key reading 5.3.1

Step 7: Ask trainees to perform the task provided in application of learning 5.3

Points to Remember

- **Steps to Repair:**
  - ✓ Setup and Baseline Monitoring
  - ✓ Simulate an Issue
  - ✓ Incident Response

- ✓ Patching and Repair
- ✓ Post-Repair Testing
- ✓ Documentation
- ✓ Review and Improve



Practical Activity 5.3.2: Performing IoT system protection checks



Notes to the trainer

- The trainer may use individual based facilitation of this activity; you are supposed to perform IoT system protection checks; it is recommended to: Avail examples and exercises on performing IoT action repairs.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:
Your organization has deployed several IoT devices, including smart cameras and environmental sensors, in an office environment. You need to perform regular protection checks to ensure these devices are secure.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to perform IoT system protection checks of the given tasks. While demonstrating, explain step-by-step performing IoT system protection checks procedures

Step 4: Give other tasks related to the activity and Asks learners to select, Performing IoT action repairs the procedures.

Step 5: Verify whether performing IoT system protection checks has well done according to the finding answers.

Step 6: Ask trainees to read key reading 5.3.2

Step 7: Ask trainees to perform the task provided in application of learning 5.3



Points to Remember

Steps to Perform Protection Checks:

- **Regular Security Audits:**

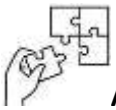
- Vulnerability Scanning:
  - ✓ Use Nessus to scan the network and identify vulnerabilities.
  - ✓ Prioritize high-risk vulnerabilities and plan remediation steps.
- Penetration Testing:
  - ✓ Use Kali Linux to perform penetration testing on the IoT network.
  - ✓ Document the vulnerabilities and how they were exploited, and create a remediation plan.

- **Access Control Review:**
  - Permission Audit:
    - ✓ Review and update access permissions for all IoT devices.
    - ✓ Remove access for former employees or roles that no longer need it.
  - Implement MFA:
    - ✓ Enable MFA for accessing the IoT management console and critical devices.
- **Network Security:**
  - Firewall Configuration:
    - ✓ Ensure the firewall is configured to block unnecessary ports and protocols.
    - ✓ Restrict incoming and outgoing traffic to what is necessary for IoT device operation.
  - Network Segmentation:
    - ✓ Create VLANs to separate IoT devices from the main office network.
- **Data Protection:**
  - Encryption Setup:
    - ✓ Ensure that all communication between IoT devices and the server is encrypted using TLS.
    - ✓ Encrypt sensitive data stored on IoT devices.
  - Integrity Checks:
    - ✓ Implement checks to verify data integrity using hash functions.
- **Compliance and Policies:**
  - Policy Review:

- ✓ Review and update security policies to include IoT device management and security.
- ✓ Ensure policies address current best practices and regulatory requirements.

- Compliance Check:
  - ✓ Conduct a compliance check to ensure all IoT devices and processes adhere to GDPR requirements.

- **User Education:**
  - Training Session:
    - ✓ Conduct training on recognizing phishing attempts and maintaining secure passwords.
    - ✓ Emphasize the importance of following security protocols for IoT devices.
  - Awareness Campaign:
    - ✓ Distribute newsletters and posters highlighting recent IoT security threats and tips for safe usage.



Application of learning 5.3

The smart city utilizes a vast network of IoT devices, including traffic management systems, environmental sensors, public safety cameras, and connected infrastructure. These systems are critical to the city's operations and must be protected against cyber threats that could disrupt services or compromise data integrity. Due to the growing sophistication of cyber threats, regular security maintenance is crucial to protect the IoT network. As a technician you are asked to perform protection checks ensuring that all devices and communication channels are secure.

Checklist for application

SN	Criteria	Indicators	Yes	No
1	Equipment, tools and materials are well prepared	1.1 Equipment is selected		
		1.2 Tools are selected		
		1.3 Materials are selected		
2	IoT system security maintenance is properly performed	2.1 IoT system protection checks are Performed		

Indicative content 5.4: Documentation of IoT Maintenance Report

Duration: 2 hrs

Practical Activity 5.4.1: Data analysing and facts recording

Notes to the trainer

- The trainer may use individual based facilitation of this activity; you are supposed to analyse data and facts recording; it is recommended to: Avail examples and exercises on performing IoT action repairs.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:
A motion sensor on your IoT network is found to be transmitting unusual amounts of data at unexpected times, indicating a potential breach or malfunction. As an IoT Technician you are asked to make Data analysing and facts recording.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to Perform IoT action repairs of the given tasks. While demonstrating, explain step-by-step Performing IoT action repairs procedures

Step 4: Give other tasks related to the activity and Asks learners to select, Performing IoT action repairs the procedures.

Step 5: Verify whether Performing IoT action repairs has well done according to the finding answers.

Step 6: Ask trainees to read key reading 5.4.1

Step 7: Ask trainees to perform the task provided in application of learning 5.4

Points to Remember

- Data analysis in IoT involves extracting meaningful insights from the vast amount of data generated by connected devices. This process is crucial for optimizing operations, improving decision-making, and identifying potential issues.
  Key steps in IoT data analysis:
  - ✓ Data Collection: Gathering data from various IoT sensors and devices.

- ✓ Data Cleaning: Ensuring data accuracy, consistency, and completeness by removing errors and inconsistencies.
- ✓ Data Transformation: Converting raw data into a suitable format for analysis.
- ✓ Data Exploration: Discovering patterns, trends, and anomalies within the data.
- ✓ Data Modeling: Creating statistical models to predict future outcomes or understand relationships between variables.
- ✓ Data Visualization: Representing data graphically to facilitate understanding and communication of insights.

Common IoT data analysis techniques:

- ✓ Descriptive statistics: Summarizing data using measures like mean, median, mode, and standard deviation.
- ✓ Exploratory data analysis (EDA): Discovering patterns and relationships within data through visualization and summary statistics.
- ✓ Predictive analytics: Forecasting future trends or outcomes based on historical data.
- ✓ Prescriptive analytics: Recommending actions based on data-driven insights.

- Facts recording in IoT refers to the systematic collection and storage of data generated by IoT devices. This data serves as the foundation for subsequent analysis and decision-making. Considerations for facts recording:
  - ✓ Data storage: Choosing appropriate storage solutions (databases, data warehouses, cloud platforms) based on data volume and structure.
  - ✓ Data format: Selecting suitable data formats (CSV, JSON, XML) for efficient storage and retrieval.
  - ✓ Data retention: Determining the necessary data retention period based on legal and business requirements.
  - ✓ Data security: Implementing robust security measures to protect sensitive data.
  - ✓ Data governance: Establishing policies and procedures for data management and access control.



Practical Activity 5.4.2: Report generation



Notes to the trainer

- The trainer may use individual based facilitation of this activity; you are supposed to analyse data and facts recording; it is recommended to: Avail examples and exercises on performing IoT action repairs.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to perform the following task:
A security camera on your IoT network is found to be transmitting unusual amounts of data at unexpected times, indicating a potential breach or malfunction. As an IoT Technician you are asked to generate a report after performing the maintenance.

Step 2: Explain the task and provide clear work instruction (Task, Time allocated)

Step 3: Demonstrate how to Perform IoT action repairs of the given tasks. While demonstrating, explain step-by-step Performing IoT action repairs procedures

Step 4: Give other tasks related to the activity and Asks learners to select, Performing IoT action repairs the procedures.

Step 5: Verify whether Performing IoT action repairs has well done according to the finding answers.

Step 6: Ask trainees to read key reading 5.4.2

Step 7: Ask trainees to perform the task provided in application of learning 5.4



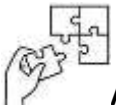
Points to Remember

- **Report Structure:** A well-structured IoT maintenance report should provide a clear overview of the maintenance activities and their impact. Consider the following structure:
  - ✓ **Executive Summary:** A concise overview of the report, highlighting key findings and recommendations.
  - ✓ **Maintenance Overview:** General information about the IoT system, including the number of devices, their criticality, and maintenance schedules.

- ✓ **Performance Metrics:** Analysis of KPIs such as MTBF, MTTR, and equipment uptime.
- ✓ **Issue Analysis:** Detailed breakdown of reported issues, including frequency, severity, and resolution time.
- ✓ **Preventive Maintenance:** Summary of preventive maintenance activities performed, including their effectiveness.
- ✓ **Corrective Maintenance:** Analysis of corrective maintenance actions, including costs and time taken.
- ✓ **Spare Parts Management:** Evaluation of spare parts inventory, usage, and replenishment strategies.
- ✓ **Cost Analysis:** Breakdown of maintenance costs, including labor, parts, and equipment.
- ✓ **Recommendations:** Suggestions for improving maintenance efficiency and reducing downtime.
- ✓ **Appendices:** Supporting data, such as maintenance logs, repair reports, and performance graphs.

- **Report Visualization:** Effective use of visuals can enhance the report's clarity and impact. Consider incorporating:
  - ✓ Graphs and charts: To represent performance metrics, issue trends, and cost analysis.
  - ✓ Tables: To summarize maintenance activities, spare parts inventory, and labor hours.
  - ✓ Maps: To visualize device locations and maintenance coverage.
- **Report Distribution and Utilization:** The maintenance report should be distributed to relevant stakeholders, including maintenance teams, management, and engineering departments. The report should be used to:
  - ✓ Identify maintenance trends: Analyze historical data to predict future maintenance needs.
  - ✓ Optimize resource allocation: Allocate maintenance resources effectively based on equipment criticality and performance.
  - ✓ Improve maintenance efficiency: Implement process improvements based on identified issues.
  - ✓ Reduce costs: Optimize spare parts inventory and maintenance schedules.

- ✓ Enhance decision-making: Provide data-driven insights for maintenance planning and strategy.



Application of learning 5.4

A smart factory utilizes an IoT system to monitor and control various manufacturing processes, including machinery performance, energy consumption, and production line efficiency. The IoT system comprises sensors, actuators, connected machines, and a central control system. Regular maintenance is essential to ensure that the IoT system operates optimally and to prevent unplanned downtime. Documenting an IoT Maintenance Report for a Smart Factory.

Checklist for application

SN	Criteria	Indicators	Yes	No
1	Data are properly Collected and Analysed	1.1 Data are collected		
		1.2 Data are Analysed		
		1.3 Temperature sensors are mounted		
2	Report is well Generated	2.1 Introduction is indicated		
		2.2 Data Analysis are summarized		
		2.3 Maintenance Activities are selected		
		2.4 Recommendations and Conclusion are indicated is indicated		



Learning outcome 5 end assessment

Written assessment

Choose the letter corresponding to the correct answer:

1. **Which IoT network technology is best suited for long-range, low-power communication?**

   A) Zigbee
   B) Wi-Fi6
   C) LoRa
   D) Bluetooth/LE

   Answer: C) LoRa

2. **What is the purpose of predictive maintenance in IoT systems**

   A) To perform regular maintenance at fixed intervals
   B) To predict when maintenance should be performed based on real-time data
   C) To replace all components periodically
   D) To monitor the physical environment of IoT devices

   Answer: B) To predict when maintenance should be performed based on real-time data

3. **Which IoT deployment level typically involves basic device installation and initial configuration?**

   A) Level-1

   B) Level-2

   C) Level-4

   D) Level-6

   Answer: A) Level-1

4. **Which of the following is NOT a common IoT testing type?**

   A) Usability

   B) Data Integrity

   C) Environmental Impact

   D) Security

Answer: C) Environmental Impact

5. **True or False:** Zigbee is a network technology suitable for long-range communication with high power consumption.

   Answer: False

6. **True or False:** Preventive maintenance involves fixing problems only when they occur. **Answer: False**

7. **True or False:** Functional testing of IoT systems involves verifying the speed, latency, and jitter of the device**.**

   Answer: True

8. **True or False:** Calibration of tools and equipment ensures that they operate with increased precision and accuracy. **Answer: True**

9. Match the following IoT network technologies with their respective range and power characteristics:

IoT network technologies	**range and power characteristics**
1. Zigbee	A) Long Range, Low Power
2. LoRa	B) Short Range, Low Power
3. Wi-Fi6	C) Long Range, High Power
4. LPWAN	D) Short Range, High Power

Answer:

1. **- B) Short Range, Low Power**
2. **- A) Long Range, Low Power**
3. **- D) Short Range, High Power**
4. **- A) Long Range, Low Power**

Practical assessment

You have been hired as an IoT maintenance technician by a manufacturing company that is managing a sophisticated IoT-based monitoring and control system within its facility. The system is designed to integrate various sensors, actuators, and controllers to monitor equipment performance, environmental conditions, and production processes in real-time. Your primary role is to ensure the system operates seamlessly and securely, focusing on software upgrades and security maintenance.

The duration of the work is 3 hours.

Resources	
Tools	Software platforms, simulation software, tape measure, multi-meter, screwdrivers, drilling machine, hammer, ladder, pliers,
Equipment	Smart Lighting System, HVAC Automation, Security Cameras and Access Control, Occupancy Monitoring Sensors and Centralized IoT Platform
Material /Consumable	Internet bundles, power extension, electricity, sensors

Checklist:

SN	Criteria	Indicators	Yes	No
1	IoT Maintenance types are well identified	1.1 Preventive maintenance is defined		
		1.2 Predictive maintenance is defined		
2	IoT Software upgrade is properly Applied	2.1 Automatic IoT Software upgrade is applied		
		2.2 Manual IoT Software upgrade is applied		
3	IoT system security maintenance is properly performed	3.1 IoT action repairs are performed		
		3.2 IoT system protection checks are performed		
4	IoT maintenance report is correctly written	4.1 Data are analysed and recorded		
		4.2 Report is generated		

END



Further information to the trainer

"Embedded Systems: Design and Applications" By Steve Heath
"Internet of Things: A Hands-On-Approach" By Arshdeep Bahga, Vijay Madisetti
"Internet of Things: Principles and Paradigms" Edited by Rajkumar Buyya, Amir Vahid Dastjerdi
"IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things"
By David Hanes, Gonzalo Salgueiro, Patrick Grossetete, and others
"IoT Systems: Architecture and Applications" By R. C. Joshi, S. S. Srivastava
"Security and Privacy in Internet of Things (IoT): Models, Algorithms, and Implementations"
Edited by Hongyu Wu, Dongxu Li
"The Internet of Things: A Guide to IoT Systems and Applications" *By Michael Miller*